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Discrete Random Variables

Random variable:
A rv, usually written as X, is a variable whose possible values are the numerical outcomes of a random phenomenon.

RV → discrete
RV → continuous

Discrete RVs:
A discrete rv is one which may take on only a countable # of distinct values. Discrete rvs are usually counts. If a variable can take only finite # of distinct values, it must be discrete.
Examples:
- the # of cars that pass through a given intersection in an hour
- the # of people in a household
- the # of goals scored in a match

Probability mass function
It is the probability distribution of a discrete rv, and provides the possible values and their associated probabilities.
It is the function p:ℝ → [0,1] defined by

$$p_X(x) = P(X = x)$$

for -∞ < x < ∞, where P is a probability measure.
p_X(x) can also be simplified as p(x).

The probabilities associated with all (hypothetical) values must be non-negative and sum up to 1,

$$\sum_x p_X(x) = 1$$

and

$$p_X(x) \geq 0.$$

E.g. I toss a fair coin three times. The rv X gives the # of heads recorded.

→ Possible values for X = {0, 1, 2, 3}

PMF:

x	0	1	2	3
P(X=x)	1/8	3/8	3/8	1/8

Cumulative distribution function
It is a function giving the probability that the rv X is less than or equal to x, for every x in the possible outcomes of X

E.g.

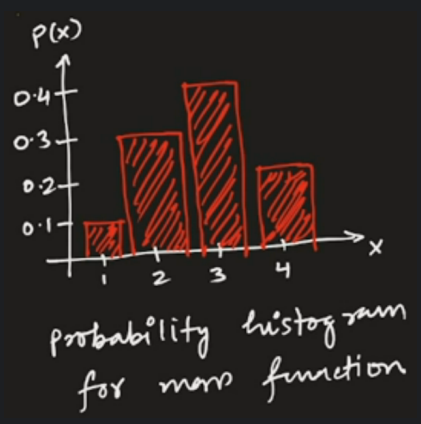
Outcome	1	2	3	4
probability	0.1	0.3	0.4	0.2

PMF

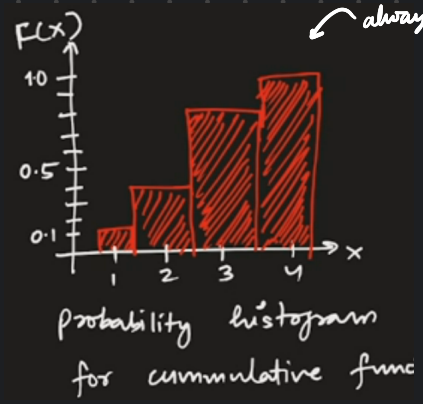
f(1) = 0.1
f(2) = 0.3
f(3) = 0.4
f(4) = 0.2

CDF

F(1) = P(X ≤ 1) = P(x=1) = 0.1
F(2) = P(X ≤ 2) = P(x=1) + P(x=2) = 0.1 + 0.3 = 0.4
F(3) = P(X ≤ 3) = f(1) + f(2) + f(3) = 0.8
F(4) = P(X ≤ 4) = f(1) + f(2) + f(3) + f(4) = 1.0



Probability histogram for mass function



Probability histogram for cummulative func

↖ always non-decreasing

Q. PMF: $f(x) = bx^3$ for $x = 1, 2, 3$. Find the value of b.

→ $f(1) + f(2) + f(3) = 1$

$b + 8b + 27b = 1$

$\Rightarrow b = 1/36$

Q. PMF:

x	0	1	2	3	4
$P(X=x)$	0	0·1	0·2	0·3	0·4

Find CDF, $F(2) = P(X \le 2)$

→ $P(X \le 2) = 0·1 + 0·2 = 0·3$

Q. PMF:

x	0	1	2
$P(X=x)$	$(2k^2+k-3)$	$(6k)$	$(8k^2+2k+3)$

$k = ?$

→ $\Sigma p_X(x) = 1$

$\Rightarrow 2k^2 + k - 3 + 6k + 8k^2 + 2k + 3 = 1$

$\Rightarrow 10k^2 + 9k - 1 = 0$

$\Rightarrow 10k^2 + 10k - k - 1 = 0$

$\Rightarrow 10k(k+1) - (k+1) = 0$

$\Rightarrow (k+1)(10k-1) = 0$

$\Rightarrow k = -1, \; k = 1/10$

x ✓

as $P(1)$ becomes < 0

Q. CDF:

x	0	1	2	3	4	5
$P(X \le x)$	0·1	0·3	0·45	0·5	0·85	1

a) Find $P(X=1)$

→ PMF:

x	0	1	2	3	4	5
$P(X=x)$	0·1	0·2	0·15	0·05	0·35	0·15

b) Find $P(1 \le X \le 4)$

→ $(0·2 + 0·15 + 0·05 + 0·35)$

$= 0·75$

Expectation

It is a measure of centre for the distribution of a rv. It is the long run mean in the sense that if more and more samples of the rv are collected, the sample mean becomes closer to the expected value.

Random variables with finitely many outcomes

Consider a random variable X with a *finite* list x_1, ..., x_k of possible outcomes, each of which (respectively) has probability p_1, ..., p_k of occurring. The expectation of X is defined as[12]

$$\mathrm{E}[X] = x_1 p_1 + x_2 p_2 + \cdots + x_k p_k.$$

Since the probabilities must satisfy $p_1 + \cdots + p_k = 1$, it is natural to interpret $\mathrm{E}[X]$ as a weighted average of the x_i values, with weights given by their probabilities p_i.

In the special case that all possible outcomes are equiprobable (that is, $p_1 = \cdots = p_k$), the weighted average is given by the standard average. In the general case, the expected value takes into account the fact that some outcomes are more likely than others.

Examples

- Let X represent the outcome of a roll of a fair six-sided die. More specifically, X will be the number of pips showing on the top face of the die after the toss. The possible values for X are 1, 2, 3, 4, 5, and 6, all of which are equally likely with a probability of $\frac{1}{6}$. The expectation of X is

$$\mathrm{E}[X] = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6} = 3.5.$$

If one rolls the die n times and computes the average (arithmetic mean) of the results, then as n grows, the average will almost surely converge to the expected value, a fact known as the strong law of large numbers.

Let X be a random variable that takes the values $a_1, a_2, \ldots, a_n$.
The expected value is given as

$$E[X] = \sum_{i=1}^{n} a_i \cdot P(X = a_i)$$

Q. You toss a coin 3 times. X : # of heads. Find E[X].

→ Say coin is fair.

x	3	2	1	0
P(X=x)	1/8	3/8	3/8	1/8

$E[X] = 3 \cdot \frac{1}{8} + 2 \cdot \frac{3}{8} + 1 \cdot \frac{3}{8} + 0 \cdot \frac{1}{8} = \frac{3}{2} = 1.5$

eg. Suppose we toss a fair coin until a heads comes up. Let X denote the number of tosses we have to make. What is the expected value of X?

→ X : 1 2 3 4 5 ⋯

$\frac{1}{2}$ $\frac{1}{4}$ $\frac{1}{8}$ $\frac{1}{16}$ $\frac{1}{32}$

$$E[X] = 1\cdot\frac{1}{2} + 2\cdot\frac{1}{4} + 3\cdot\frac{1}{8} + \cdots$$

$$S = 1\cdot\frac{1}{2} + 2\cdot\frac{1}{4} + 3\cdot\frac{1}{8} + 4\cdot\frac{1}{16} + \cdots$$

$$S/2 = \quad 1\cdot\frac{1}{4} + 2\cdot\frac{1}{8} + 3\cdot\frac{1}{16} + \cdots$$

$$S/2 = 1\cdot 1/2 + 1\cdot 1/4 + 1\cdot 1/8 + \cdots$$

$$S/2 = 1$$

$$S = 2$$

$$E[X] = 2$$

The sum of 2 random variables is a random variable as well.

Example:
We flip a coin. X = 1 if heads, 0 if tails.
Then we roll a die, Y = value on top.
What is X+Y?

$X : \Omega = \{0, 1\}$

$Y : \Omega = \{1, 2, 3, 4, 5, 6\}$

X \ Y	1	2	3	4	5	6
0	1/12	1/12	1/12	1/12	1/12	1/12
1	1/12	1/12	1/12	1/12	1/12	1/12

pmf:

$x+y$	1	2	3	4	5	6	7
$P(X+Y=x+y)$	1/12	2/12	2/12	2/12	2/12	2/12	1/12

$$E[X+Y] = (1+7)\frac{1}{12} + (2+3+4+5+6)\cdot\frac{2}{12}$$

$$= 8/12 + \frac{40}{12} = 12/3 = 4 = 0.5 + 3.5$$

$$= E[X] + E[Y]$$

Linearity of expectation

$E[X+Y] = E[X] + E[Y]$

Proof:

$X: \Omega = \{x_1, x_2, \dots\}$

$Y: \Omega = \{y_1, y_2, \dots\}$

$X+Y = \{x_1+y_1,\ x_1+y_2, \dots, x_2+y_1, \dots\}$

$= \{x_i + y_j \ \forall i, j\}$

$$E[X+Y] = \sum_i \sum_j (x_i + x_j) \cdot P(X=x_i, Y=y_j)$$

$$= \sum_i \sum_j x_i \cdot P(X=x_i, Y=y_j) + \sum_i \sum_j x_j \cdot P(X=x_i, Y=y_j)$$

$$= \sum_i x_i P(X=x_j) + \sum_j x_j P(X=x_j)$$

$$= E[X] + E[Y]$$

$E[aX+b] = aE[X] + b$

Proof:

$X: \Omega = \{\underset{p_1}{x_1}, \underset{p_2}{x_2}, \dots\}$

$aX+b: \Omega = \{\underset{p_1}{ax_1+b}, \underset{p_2}{ax_2+b}, \dots\}$

$$E[aX+b] = \Sigma (ax_i + b) \cdot P(X=x_i)$$

$$= a\Sigma x_i \cdot P(X=x_i) + b\Sigma P(X=x_i)$$

$$= a\,E[X] + b$$

Q. A bucket contains 4 different colored balls. Select 4 balls with replacement. What is the expected # of distinct colored balls you will select?

Q. A fair coin flip experiment is carried 6 times. What's the expected # of heads?

→ X : # of heads

x	0	1	2	3	4	5	6
$f(x)$	$1/64$	$6/64$	$15/64$	$\frac{{}^6C_3}{64}$	$\frac{15}{64}$	$\frac{6}{64}$	$\frac{1}{64}$

$$E(X) = 0 \cdot \frac{1}{64} + 1 \cdot \frac{6}{64} + \dots + 6 \cdot \frac{1}{64}$$

X_1 : Heads appears on the first place. → 1 ($p = 1/2$), 0 ($p = 1/2$)

X_2 : heads on 2nd place → 1, 0

$\vdots$

X_6

Indicator variable.

$X = X_1 + X_2 + X_3 + X_4 + X_5 + X_6$ [will give # of heads]

$$E[X] = E[X_1] + E[X_2] + \dots + E[X_6]$$

$$= 1/2 + 1/2 + \dots + 1/2$$

$$= 3$$

Q. Consider all the permutations of set { a, b, c }. Let Y denote the # of fixed points in any random permutation. What is E[Y]?

→ Y_1 : a is at correct position → 1 ($2/6$), 0 ($4/6$)

{abc acb}

$$E[Y_1] = 1 \cdot \frac{2}{6} + 0 \cdot \frac{4}{6} = \frac{2}{6}$$

Y_2 : b is at correct position $2/6$

Y_3 : c $2/6$

$$E[Y] = E[Y_1] + E[Y_2] + E[Y_3]$$

$$= \frac{2}{6} \cdot 3 = 1$$

Q. A bucket contains 4 different colored balls. Select 4 balls with replacement. What is the expected # of distinct colored balls you will select?

→ $c_1\ c_2\ c_3\ c_4$

X: # of distinct colored balls.

X_1: we include c_1 ← 1 if yes, 0 if no

$\frac{3^4}{4^4} = \frac{81}{256}$ ← $1 - \frac{81}{256} = \frac{175}{256}$

$E[X_1] = 1 \cdot \frac{175}{256} + 0 \cdot \frac{81}{256} = \frac{175}{256}$ Ly, X_2, X_3, X_4.

$X = X_1 + X_2 + X_3 + X_4$

$E[X] = E[X_1] + E[X_2] + E[X_3] + E[X_4]$

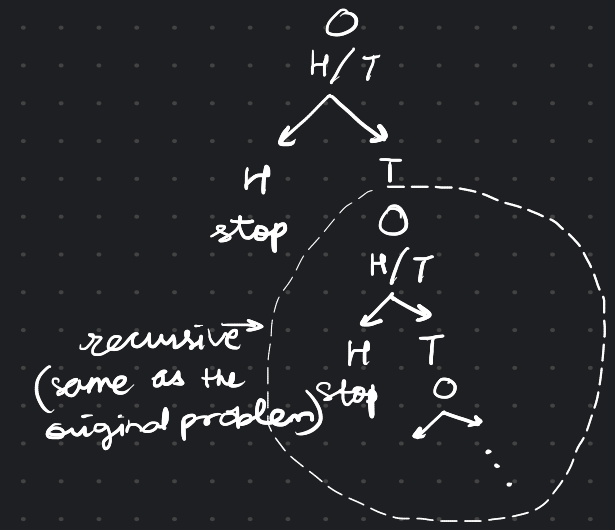
$= 4 \cdot \frac{175}{256} = \frac{175}{64}$.

Expectation as monte carlo chain:

Q. Expected # of coin flips to get a head?

→

$$E[X] = \underbrace{(1 \cdot P(H))}_{\text{\# of moves if you get a heads}} + \underbrace{(1 + E[X]) \cdot P(T)}_{\text{\# of moves if you get a tails}}$$



$E = \frac{1}{2} + (1 + E)\frac{1}{2}$

$E = \frac{1}{2} + \frac{1}{2} + \frac{E}{2}$

$E = 2$

Q. Expected # of coin flips required to get 2 consecutive heads?

→ $E = 2 \cdot P(HH) + [2 + E]\,P(HT) + [1 + E]\,P(T)$

$E = 2 \cdot \frac{1}{4} + (2 + E)\frac{1}{4} + (1 + E)\frac{1}{2}$

$E = \frac{1}{2} + \frac{1}{2} + \frac{E}{4} + \frac{1}{2} + \frac{E}{2}$

$\frac{E}{4} = \frac{3}{2}$ $\quad E = 6$

Q. Expected # of coin flips required to get n consecutive heads?

Q. Candidates are appearing for an interview one after the other. Probability of each candidate getting selected is 0.16. What is the expected # of candidates that you'll need to interview to make sure that you select somebody?

→ $E = 1 \cdot (0.16) + (1 + E)(0.84)$

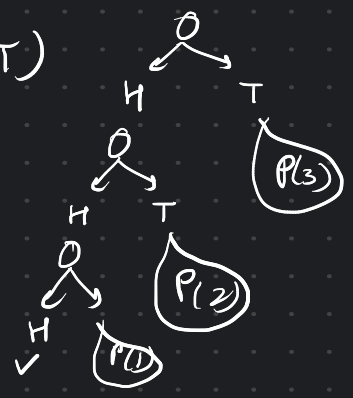
$E = 0.16 + 0.84 + 0.84E$

$0.16E = 1$

$E = \frac{1}{0.16} = 6.25$

Q. Expected # of coin flips reqd to get atleast 3 heads?

$$E[X_n] = \left(1 + E[X_{n-1}]\right) \cdot P(H) + \left(1 + E[X_n]\right) \cdot P(T)$$



$$E(X_n) = \frac{1}{2} + \frac{1}{2} E[X_{n-1}] + \frac{1}{2} + \frac{1}{2} E[X_n]$$

$$\frac{1}{2} E[X_n] = \frac{1}{2} E[X_{n-1}] + 1$$

$$E[X_n] = E[X_{n-1}] + 2 \qquad \text{base case } E[1] = 2$$

$$E[3] = E[2] + 2 = E[1] + 2 + 2 = \underline{6}$$

$$E[X_n] = 2n$$

Conditional probability distribution

If X and Y are random variables with joint pmf P(X, Y), then the conditional probability mass function of x, given Y=y is given as

$$P_{X|y}(x|y) = \frac{P(X=x \text{ and } Y=y)}{P(Y=y)} = \frac{P(x,y)}{P_Y(y)}, \quad P_Y(y) > 0$$

eg: We toss a fair coin three times and record the sequence of heads (h) and tails (t). We let random variable X denote the number of heads obtained. We also let random variable Y denote the winnings earned in a single play of a game with following rules:

- player wins $1 if first h occurs on first toss
- player wins $2 if first h occurs on second toss
- player wins $3 if first h occurs on third toss
- player loses $1 if no h occur.

→

x	0	1	2	3
$P_X(x)$	1/8	3/8	3/8	1/8

$$E[X] = \frac{3}{8}\cdot 1 + \frac{3}{8}\cdot 2 + \frac{1}{8}\cdot 3 = 1.5$$

HHH, HHT, HTH, THH, HTT, THT, TTH, TTT

y	+1	+2	+3	−1
$P_Y(y)$	4/8	2/8	1/8	1/8

$$E[Y] = \frac{4}{8}\cdot 1 + \frac{2}{8}\cdot 2 + \frac{1}{8}\cdot 3 + \frac{1}{8}(-1)$$

$$= \frac{10}{8} = \frac{5}{4} = 1.25$$

joint PMF:

y \ x	0	1	2	3
−1	1/8	0	0	0
1	0	1/8	1/4	1/8
2	0	1/8	1/8	0
3	0	1/8	0	0

← 1/2 = P(Y=1)

	X	Y
HHH	3	1
HHT	2	1
HTH	2	1
THH	2	2
HTT	1	1
THT	1	2
TTH	1	3
TTT	0	−1

conditional PMF of X given $Y = y$.

x	$P_{X\|Y}(x\|-1)$
0	1 ← $\frac{P(X=0, Y=-1)}{P(Y=-1)} = \frac{1/8}{1/8} = 1$
1	0
1	0
1	0

x	$P_{X\|Y}(x\|1)$
0	0
1	1/4
1	1/2
1	1/4

Conditional expectation

If F is any event and X is a random variable with sample space $\Omega = \{x_1, x_2, \ldots\}$ then the conditional expectation is given as

$$E[X|F] = \sum_i x_i \cdot P(X = x_i | F)$$

e.g.

x	$P_{X\|Y}(x\|-1)$ ↓	$P_{X\|Y}(x\|1)$	$P_{X\|Y}(x\|2)$	$P_{X\|Y}(x\|3)$
0	1	0	0	0
1	0	1/4	1/2	1
2	0	1/2	1/2	0
3	0	1/4	0	0
	1	1	1	1

$E[X|Y=-1] = 0\cdot 1 + 1\cdot 0 + 2\cdot 0 + 3\cdot 0 = 0$

$E[X|Y=1] = 0\cdot 0 + 1\cdot \frac{1}{4} + 2\cdot \frac{1}{2} + 3\cdot \frac{1}{4} = 2$

$E[X|Y=2] = 0\cdot 0 + 1\cdot \frac{1}{2} + 2\cdot \frac{1}{2} + 3\cdot 0 = 1.5$

$E[X|Y=3] = 1$

Theorem:

Let X be a r.v. with sample space Ω. If $F_1, F_2, \ldots, F_n$ are events s.t. $F_i \cap F_j = \phi \ \forall i, j$ and $\Omega = \bigcup_i F_i$, then

$$E[X] = \sum_i E[X|F_i] \cdot P(F_i)$$

Proof:

$$\text{RHS} = \sum_i E[X|F_i] \cdot P(F_i)$$

$$= \sum_i \sum_j x_j P(X = x_j | F_i) \cdot P(F_i)$$

$$= \sum_i \sum_j x_j P(X = x_j, F_i)$$

$$= \sum_j x_j \sum_i P(X = x_j, F_i)$$

$$= \sum_j x_j P(X = x_j)$$

$$= E[X]$$

Same as

$$E[X] = \sum_y E[X|Y=y] \cdot P(Y=y)$$

If $Y=y$ are disjoint and mutually exclusive.

In the given example,

$E[X|Y=-1] = 0$

$E[X|Y=1] = 2$

$E[X|Y=2] = 1.5$

$E[X|Y=3] = 1$

$$\therefore E[X] = \sum_i E[X|Y=y_i] \cdot P(Y=y_i)$$

$$= (0)\frac{1}{8} + (2)\frac{1}{2} + (1.5)\cdot\frac{1}{4} + (1)\cdot\frac{1}{8}$$

$$= \frac{12}{8} = \frac{3}{2} = 1.5$$ ← same as calculated previously.

Theorem:

Let us denote by $E[X|Y]$ that func^ of the random variable Y whose value at $Y=y$ is $E[X|Y=y]$.

Note that $E[X|Y]$ itself is a r.v.

$$E[E[X|Y]] = E[X].$$

whose sample space has (in the given example)
- $E[X|Y=1]$ ← $P(Y=1)$
- $E[X|Y=2]$ ← $P(Y=2)$
- $E[X|Y=3]$
- $E[X|Y=-1]$

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eg: Let X, Y be random variables with joint pmf:

$$P_{X,Y}(x,y) = \begin{cases} 1/4, & \text{if } x=2,\ y=2 \\ 1/4, & \text{if } x=2,\ y=0 \\ 1/4, & \text{if } x=1 \text{ and } y=2 \\ 1/4, & \text{if } x=0,\ y=2 \\ 0, & \text{otherwise} \end{cases}$$

$S_X = \{0, 1, 2\}$

$S_Y = \{0, 2\}$

what is the conditional expectation of X given $Y=2$?

Practice questions

Two coins are to be flipped. The first coin will land on heads with probability .6, the second with probability .7. Assume that the results of the flips are independent, and let X equal the total number of heads that result.

(a) Find $P\{X = 1\}$.

(b) Determine $E[X]$.

	1st	2nd
H	0.6	0.7
T	0.4	0.3

X: # of heads

a) $P(X=1) = (0.6)(0.3) + (0.4)(0.7) = 0.46$

b) $E[X] = E[X_1] + E[X_2] = 0.6 + 0.7 = 1.3$

(Indicator variables)

One of the numbers 1 through 10 is randomly chosen. You are to try to guess the number chosen by asking questions with "yes–no" answers. Compute the expected number of questions you will need to ask in each of the following two cases:

(a) Your ith question is to be "Is it i?" $i = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10$.

(b) With each question you try to eliminate one-half of the remaining numbers, as nearly as possible.

(a) we'll ask "is it 1?", "is it 2?", ...

X: # of questions asked. $x: \{1, 2, \cdots, 10\}$

x	1	2	3	...
$P(X=x)$	1/10	1/10	1/10	

$E[x] = \frac{1}{10}(1+2+..+10) = 5.5$

(b) we do smth. like binary search.

1 2 3 4 5 6 7 8 9 10

we'll ask if chosen # $\geq$ mid.

search space would be

1 2 3 4 5 6 7 8 9 10

1 2 | 3 4 5 | 6 7 | 8 9 10

1 | 2 | 3 | 4 5 | 6 | 7 | 8 | 9 10

4 | 5 | 9 | 10

$X: \{3, 4\}$ pr. $\frac{6}{10}$ $\frac{4}{10}$

$E[X] = 3 \cdot \frac{6}{10} + 4 \cdot \frac{4}{10} = 3.4$

A player throws a fair die and simultaneously flips a fair coin. If the coin lands heads, then she wins twice, and if tails, then one-half of the value that appears on the die. Determine her expected winnings.

→ Heads → 2 * value of die
Tails → $\frac{1}{2}$ * value of die.

$$\frac{1}{2}\left[\frac{1}{6}\left[\frac{1+2+3+4+5+6}{2}\right]\right] + \frac{1}{2}\left[\frac{1}{6}\left[1+2+3+4+5+6\right]\times 2\right]$$

$$= \frac{1}{2}\cdot\frac{1}{6}\cdot\left[\frac{21}{2} + 21\cdot 2\right]$$

$$= \frac{21}{2\cdot 6}\left[\frac{5}{2}\right] = \frac{105}{24} = 4.375$$

$D: \{1, 2, 3, 4, 5, 6\}$ W: winnings

$C: \{H, T\}$

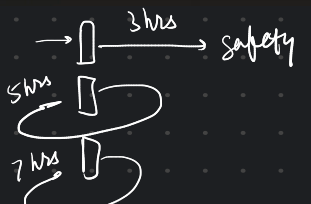
$$E[W|C=H] = E[2D] = 2\cdot\frac{1}{6} + 4\cdot\frac{1}{6} + 6\cdot\frac{1}{6} + 8\cdot\frac{1}{6} + 10\cdot\frac{1}{6} + 12\cdot\frac{1}{6} = 7$$

$$E[W|C=T] = E\left[\frac{D}{2}\right] = \frac{1}{2}\cdot\frac{1}{6} + \frac{2}{2}\cdot\frac{1}{6} + \frac{3}{2}\cdot\frac{1}{6} + \frac{4}{2}\cdot\frac{1}{6} + \frac{5}{2}\cdot\frac{1}{6} + \frac{6}{2}\cdot\frac{1}{6} = \frac{7}{4}$$

$$E[W] = E[W|C=H]\cdot P(C=H) + E[W|C=T]\cdot P(C=T)$$

$$= 7\cdot\frac{1}{2} + \frac{7}{4}\cdot\frac{1}{2} = 4.375$$

A miner is trapped in a mine containing 3 doors. The first door leads to a tunnel that will take him to safety after 3 hours of travel. The second door leads to a tunnel that will return him to the mine after 5 hours of travel. The third door leads to a tunnel that will return him to the mine after 7 hours. If we assume that the miner is at all times equally likely to choose any one of the doors, what is the expected length of time until he reaches safety?



$$E[X] = \frac{1}{3}\cdot 3 + \frac{1}{3}[5 + E(X)] + \frac{1}{3}[7 + E(X)]$$

$$E = 1 + \frac{5}{3} + \frac{E}{3} + \frac{7}{3} + \frac{E}{3}$$

$$\frac{E}{3} = \frac{15}{3} \qquad E = 15$$

Joint PMF

If discrete random variables X and Y are defined on the sample space S, then their joint PMF is given by

$$P(x, y) = P(X=x \text{ and } Y=y),$$

where (x, y) is a pair of possible values for the pair of random variables (X, Y). P(x, y) satisfies the conditions: $0 \leq P(x, y) \leq 1$, $\sum\sum_{(x,y)} P(x, y) = 1$

Joint CDF

We can obtain the joint CDF of X and Y by summing the joint PMF as follows:

$$F(x, y) = P(X \leq x \text{ and } Y \leq y) = \sum_{x_i \leq x} \sum_{y_i \leq y} P(x_i, y_i)$$

Marginal PMF

eg: We toss a fair coin three times and record the sequence of heads (h) and tails (t). We let random variable X denote the number of heads obtained. We also let random variable Y denote the winnings earned in a single play of a game with following rules:

- player wins \$1 if first h occurs on first toss
- player wins \$2 if first h occurs on second toss
- player wins \$3 if first h occurs on third toss
- player loses \$1 if no h occur.

→ HHH, HHT, HTH, HTT, THH, THT, TTH, TTT

X: 3 2 2 1 2 1 1 0

Y: 1 1 1 1 2 2 3 -1

$X: \Omega = \{0, 1, 2, 3\}$ $Y: \Omega = \{-1, 1, 2, 3\}$

$P(X=0 \text{ and } Y=-1) = \frac{1}{8}$

$P(x, y)$:

y \ x	0	1	2	3	
-1	1/8	0	0	0	1/8
1	0	1/8	2/8	1/8	4/8
2	0	1/8	1/8	0	2/8
3	0	1/8	0	0	1/8
	1/8	3/8	3/8	1/8	

Joint PMF

Joint CDF:

y \ x	0	1	2	3
-1	1/8	1/8	1/8	1/8
1	1/8	2/8	4/8	5/8
2	1/8	3/8	6/8	7/8
3	1/8	4/8	7/8	1

marginal PMF:

y	-1	1	2	3
P(Y=y)	1/8	4/8	2/8	1/8

x	0	1	2	3
P(X=x)	1/8	3/8	3/8	1/8

Independent random variables

Discrete random variables X1, X2, ..., Xn are independent if the joint PMF factors into a product of the marginal PMFs.

$$P(x_1, x_2, \ldots, x_n) = P(x_1) \cdot P(x_2) \cdots P(x_n)$$

Theorem:

If X and Y are independent random variables, then $E[XY] = E[X] \cdot E[Y]$

Expectation of a function of a random variable

this is called the support vector of the random variable.

E.g. Let X denote a random variable that takes the values {-1, 0, 1} with respective probabilities P(X=-1) = 0.2, P(X=0) = 0.5, and P(X=1) = 0.3.

→ $E[X] = (-1)(0.2) + (0)(0.5) + (1)(0.3) = 0.1$

$Y = X^2$

$Y : \Omega = \{0, 1\}$

$x=0$ ↑ $P = 0.5$

$x=-1$ or $x=1$ ↑ $0.2 + 0.3 = 0.5$

y	0	1
$P(Y=y)$	0.5	0.5

$E[Y] = E[X^2] = (0.5)(0) + (0.5)(1)$

$= 0.5$

or $E[X^2] = x_1^2 P(x_1) + x_2^2 P(x_2) + x_3^2 P(x_3)$

$= 1(0.2) + 0(0.5) + 1(0.3)$

$= 0.5$

Proposition

If X is a discrete rv that takes the values xi, i>=1 with respective probabilities P(xi), then for any real-valued function g,

$$E[g(X)] = \sum_i g(x_i) \cdot P(x_i)$$

Expectation of functions of jointly distributed rvs

Suppose X and Y are jointly distributed discrete rvs with joint PMF P(x, y).
If g(X, Y) is a function of these rvs, then the expected value is given as

$$E[g(X,Y)] = \sum_x \sum_y g(x,y) \cdot P(x,y)$$

Q. Calculate $E[XY]$

Y \ X	0	1	2	3
-1	1/8	0	0	0
1	0	1/8	2/8	1/8
2	0	1/8	1/8	0
3	0	1/8	0	0

Joint PMF

$E[XY] = (0)(-1)\frac{1}{8} + (1)(1)(1/8) + (1)(2)(1/8) + (1)(3)(1/8)$

$+ (2)(1)(2/8) + (2)(2)(1/8) + (3)(1)(1/8)$

$= 17/8$

Proof: $E[XY] = E[X] \cdot E[Y]$

$$E[XY] = \sum_x \sum_y x \cdot y \cdot p(x, y)$$

$$= \sum_x \sum_y x \cdot y \cdot p(x)\, p(y) \quad \because X \text{ and } Y \text{ are independent}$$

$$= \sum_x x\, p(x) \sum_y y\, p(y)$$

$$= E[X] \cdot E[Y]$$

Variance of discrete random variables:

Consider an example. There are 3 random variables: X, Y, and Z

$X = 0, \quad p = 1$

$Y = -1, \quad p = 1/2$
$+1, \quad p = 1/2$

All of these have an expected value of 0.
But Z is spread much farther from the expected value.
Variance gives us the spread from the mean.

$Z = -100, \quad p = 1/3$
$0, \quad p = 1/3$
$100, \quad p = 1/3$

$$\boxed{Var[x] = E[(x-\mu)^2]}$$

- $Var[X] = E[(X-\mu)^2] = E[X^2] - (E[X])^2$

Proof:

$$E[(X-\mu)^2] = \sum_x (x-\mu)^2 \cdot p(x)$$
$$= \sum_x x^2 \cdot p(x) - 2\mu \sum_x x \cdot p(x) + \mu^2 \sum_x p(x)$$
$$= E[X^2] - 2\mu \cdot \mu + \mu^2$$
$$= E[X^2] - \mu^2$$
$$= E[X^2] - (E[X])^2$$

Standard deviation of a D.R.V.
The standard deviation of X, denoted by D(X) is $D(X) = \sqrt{V(X)}$ or $\sigma = \sqrt{Var(X)}$

Q. Consider 1 roll of a die. Let X be the # that comes up. What is var(X)?

→

$X = \{1, 2, 3, 4, 5, 6\}$
$\uparrow$
$p = 1/6$

$E[X] = \frac{1}{6}(1+2+\cdots+6) = \frac{7}{2}$ $\qquad$ $E[X^2] = \frac{1}{6}[1^2+2^2+3^2+4^2+5^2+6^2] = \frac{91}{6}$

$Var(X) = E[X^2] - (E[X])^2 = \frac{91}{6} - \frac{49}{4} = \frac{182-147}{12} = \frac{35}{12}$

X	$p(x=x)$	$x-\mu$	$(x-\mu)^2$
1	1/6	-2.5	6.25
2	1/6	-1.5	2.25
3	1/6	-0.5	0.25
4	1/6	0.5	0.25
5	1/6	1.5	2.25
6	1/6	2.5	6.25

$E[x] = 3.5 = \mu$

$E[(x-\mu)^2] = 6.25 \times \frac{1}{6} + \cdots + 6.25 \times \frac{1}{6}$
$= 35/12$

$E[(x-\mu)^2] = \frac{1}{6}[12.5 + 4.5 + 0.5]$
$= \frac{1}{6}[17.5]$
$= \boxed{\frac{35}{12}} \approx 2.91$

Identity: $\text{Var}(aX+b) = a^2 \text{Var}[X]$

Proof: $\text{Var}[aX+b] = E[(aX+b)^2] - (E[aX+b])^2$

$= E[a^2X^2 + 2abX + b^2] - (aE[X] + b)^2$

$= a^2 E[X^2] + 2ab E[X] + b^2 - a^2(E[X])^2 - 2ab E[X] - b^2$

$= a^2 [E[X^2] - (E[X])^2]$

$= a^2 \cdot \text{Var}[X]$

$\text{Var}(aX+b) = E[(aX+b - a\mu - b)^2]$

$= E[a^2 \cdot (X-\mu)^2]$

$= a^2 \cdot E[(X-\mu)^2]$

$= a^2 \cdot \text{Var}(X)$

Theorem:
If X and Y are independent rvs, then $V(X+Y) = V(X) + V(Y)$

Proof: $\text{Var}[X+Y] = E[(X+Y)^2] - (E[X+Y])^2$

$= E[X^2] + 2E[XY] + E[Y^2] - (E[X])^2 - (E[Y])^2 - 2E[X]E[Y]$

$= E[X^2] - (E[X])^2 + E[Y^2] - (E[Y])^2$

$= \text{Var}(X) + \text{Var}(Y)$

Let X1, X2, X3, ..., Xn be independent random variables with
$E[X_i] = \mu$ and $\text{Var}[X_i] = \sigma^2$ $\forall i$

Let Sn = X1 + X2 + ... + Xn be the sum random variable and An = Sn/n is the average rv.
Then

$E[S_n] = n \cdot \mu$

$\text{Var}[S_n] = n\sigma^2, \quad \sigma[S_n] = \sqrt{n}\,\sigma$

$E[A_n] = \mu$

$\text{Var}[A_n] = \frac{\sigma^2}{n}, \quad \sigma[A_n] = \frac{\sigma}{\sqrt{n}}$

proof

$\text{Var}\left[\frac{S_n}{n}\right] = \frac{1}{n^2}\text{Var}[S_n]$

$= \frac{1}{n^2} n\sigma$

$= \frac{\sigma}{n}$

Conditional variance

We know that $\mu_{X|y} = E[X|Y=y] = \sum_x x\, P(X=x|Y=y)$

The conditional variance of X given Y is

$$\sigma^2_{X|y} = E\left[(X-\mu_{X|y})^2 \,|\, y\right] = \sum_x (x-\mu_{X|y})^2 \cdot P(X=x|Y=y)$$

$$\sigma^2_{X|y} = E[X^2|y] - \mu^2_{X|y} = \sum_x x^2 \cdot P(x|y) - \mu^2_{X|y}$$

$$Var[X|y] = E[(X-E[X|Y])^2|Y]$$

---

e.g. Let X be a drv with support S1 = {0, 1}, and let Y be a drv with support S2 = {0, 1, 2}. Suppose the joint PMF of X and Y is given as f(x, y) as follows:

f(x,y)	Y=0	Y=1	Y=2
X=0	1/8	2/8	1/8
X=1	2/8	1/8	1/8

Find $\sigma^2_{Y|X=0}$ or $Var[Y|X=0]$

→ $Y|X=0$

conditional PMF:

y	0	1	2
$P(Y=y\|X=0)$	1/4	2/4	1/4

$E[Y|X=0] = \frac{1}{4}\cdot 0 + \frac{2}{4}\cdot 1 + \frac{1}{4}\cdot 2 = 1$

$E[Y^2|X=0] = \frac{1}{4}\cdot 0 + \frac{2}{4}\cdot 1 + \frac{1}{4}\cdot 4 = \frac{6}{4} = 1.5$

$Var[Y|X=0] = E[Y^2|X=0] - (E[Y|X=0])^2$

$= 1.5 - (1)^2 = 0.5$

Or

$$\sigma^2_{Y|X=0} = E\left[(Y-\mu_{Y|0})^2 \,|\, X=0\right]$$

$$= (0-1)^2 \cdot P(Y=0|X=0) + (1-1)^2 \cdot P(Y=1|X=0) + (2-1)^2 \cdot P(Y=1|X=0)$$

$$= 1\times\tfrac{1}{4} + 0\times\tfrac{1}{2} + 1\times\tfrac{1}{4} = \boxed{\tfrac{1}{2}}$$

Consider 2 rvs X and Y with their joint PMFs given as

	Y=0	Y=1
X=0	1/5	2/5
X=1	2/5	0

Let Z = E[X|Y]

↳ $\Omega = \{E[X|Y=0], E[X|Y=1]\}$

Q. Find the marginal PMFs of X and Y

→ marginal PMF(X):

x	0	1
P(X=x)	3/5	2/5

PMF(Y):

y	0	1
P(Y=y)	3/5	2/5

Q. Find the conditional PMFs of X given Y=0 and Y=1.

→ given Y=0:

P(Y=0) = 3/5

x	0	1
P(X=x|Y=0)	1/3	2/3

given Y=1:

P(Y=1) = 2/5

x	0	1
P(X=x|Y=1)	1	0

Q. Find the PMF of Z.

→ Z = E[X|Y]

x	0	1
P(X=x|Y=0)	1/3	2/3

$E[X|Y=0] = 0 \cdot 1/3 + 1 \cdot 2/3 = 2/3$

x	0	1
P(X=x|Y=1)	1	0

$E[X|Y=1] = 0$

PMF of Z, i.e. PMF of E[X|Y]

z	2/3	0
P(Z=z)	3/5	2/5

← probabilities of Y=0 and Y=1.

$E[Z] = \frac{2}{3} \cdot \frac{3}{5} + 0 \cdot \frac{2}{5} = \frac{2}{5}$

↳ E[E[X|Y]]

$= \sum_y E[X|Y=y] \cdot P(Y=y)$

$E[X] = 0 \times 3/5 + 1 \times 2/5$ (X=0, X=1)

$= 2/5$

$E[E[X|Y]] = E[X]$ → Law of iterated expectations

Q. Find var(Z)

$E[Z^2] = \frac{4}{9} \cdot \frac{3}{5} + 0 \cdot \frac{2}{5} = \frac{4}{15}$

$Var(Z) = E[Z^2] - (E[Z])^2 = \frac{4}{15} - \left(\frac{2}{5}\right)^2 = \frac{4}{15} - \frac{4}{25} = \frac{20-12}{75} = \frac{8}{75}$

Q. Let $V = Var[X|Y]$
Find PMF of V
Find $E[V]$
Verify that $Var[X] = E[V] + Var[E]$

→ $V = Var[X|Y]$ will take values $Var[X|Y=0]$ and $Var[X|Y=1]$.

x	0	1
$P(X=x\|Y=0)$	1/3	2/3

x	0	1
$P(X=x\|Y=1)$	1	0

$E[X|Y=0] = 2/3$ $\quad$ $E[X|Y=1] = 0$

$E[X^2|Y=0] = 2/3$ $\quad$ $E[X^2|Y=1] = 0$

$Var[X|Y=0] = \frac{2}{3} - \frac{4}{9} = \frac{2}{9}$ $\quad$ $Var[X|Y=1] = 0 - 0^2 = 0$

PMF of V:

v	$2/9$	0
$P(V=v)$	3/5	2/5

$E[V] = \frac{2}{9}\cdot\frac{3}{5} + 0\cdot\frac{2}{5} = \frac{2}{15}$

marginal PMF(X):

x	0	1
$P(X=x)$	3/5	2/5

To verify: $Var[X] = E[V] + Var[E]$

RHS $= E[V] + Var[E]$
$= \frac{2}{15} + \frac{8}{75}$
$= \frac{18}{75} = \frac{6}{25}$

LHS $= Var[X]$
$= 1^2\cdot\frac{2}{5} - \left(1\cdot\frac{2}{5}\right)^2$
$= \frac{2}{5} - \frac{4}{25} = \frac{6}{25}$

$\therefore Var[X] = E[V] + Var[E]$ ✓✓

NOTE: $Var[X] = E[Var[X|Y]] + Var[E[X|Y]]$
↳ law of total variance/ variance decomposition

Proof: $Var[X|Y] = E[X^2|Y] - (E[X|Y])^2$

$E[Var[X|Y]] = \underline{E[E(X^2|Y)]} - E\left[(E[X|Y])^2\right]$ ——①

$E[E(X^2|Y)] = E[X^2]$ (law of iterated expectations)

$Var[E[X|Y]] = E\left[(E[X|Y])^2\right] - \left(E[E[X|Y]]\right)^2$

$= E\left[(E[X|Y])^2\right] - (E[X])^2$ ——②

①+②, $E[Var[X|Y]] + Var[E[X|Y]] = E[X^2] - (E[X])^2$

$E[Var[X|Y]] + Var[E[X|Y]] = Var[X]$

Covariance of drv

Let X and Y be random variables with means μ_X and μ_Y.

$$\begin{aligned} \text{cov}(X,Y) &= E[(X-\mu_X)(Y-\mu_Y)] \\ &= E[(X-E[X])(Y-E[Y])] \\ &= \sum_x \sum_y (x-\mu_X)(y-\mu_Y)\cdot P(x,y) \end{aligned}$$

eg: Suppose X and Y have the joint pmf as follows:

P(x,y)	Y = 1	2	3	P(X=x)
X = 1	0.25	0.25	0	0.50
2	0	0.25	0.25	0.50
P(Y=y)	0.25	0.50	0.25	

- what is the covariance of X and Y?

→ $E[X] = 1\cdot(0.5) + 2(0.5) = 1.5$

$E[Y] = 1\cdot\frac{1}{4} + 2\cdot\frac{1}{2} + 3\cdot\frac{1}{4} = 2$

$$\begin{aligned} \text{cov}(X,Y) &= E[(X-\mu_X)(Y-\mu_Y)] \\ &= \sum_x \sum_y (x-\mu_X)(y-\mu_Y)\cdot P(x,y) \\ &= (1-1.5)(1-2)(0.25) + (1-1.5)(2-2)(0.25) + (1-1.5)(3-2)(0) \\ &\quad + (2-1.5)(1-2)(0) + (2-1.5)(2-2)(0.25) + (2-1.5)(3-2)(0.25) \\ &= \frac{1}{4} \end{aligned}$$

$$\begin{aligned} \text{cov}(X,Y) &= E[(X-E(X))\cdot(Y-E[Y])] \\ &= E[XY - XE[Y] - YE[X] + E[X]E[Y]] \\ &= E[XY] - E[Y]E[X] - E[X]E[Y] + E[X]E[Y] \\ &= E[XY] - E[X]E[Y] \end{aligned}$$

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

If X, Y are independent, $E[XY] = E[X]E[Y]$

$\therefore \text{cov}(X,Y) = 0$

If X and Y are independent, their covariance is 0.

NOTE: cov(X,Y) could be 0 even if X, Y are dependent.

→

eg: X, Y are discrete random variables such that:

X: $P(X=0) = P(X=-1) = P(X=1) = 1/3$

Y is defined as: $Y = \begin{cases} 0, & X \neq 0 \\ 1, & X = 0 \end{cases}$

Find the cov of X and Y

→

x	0	1	-1
$P(X=x)$	1/3	1/3	1/3

y	0	1
$P(Y=y)$	2/3	1/3

Joint PMF:

$P(x,y)$	y: 0	1
x: -1	1/3	0
0	0	1/3
1	1/3	0

$P(X=0, Y=1) = P(X=0)$

$P(X\neq 0, Y=0) = P(X\neq 0)$

$$\begin{aligned} \text{cov}(X,Y) &= E[XY] - E[X]E[Y] \\ &= 0 - 0 \cdot \tfrac{1}{3} \\ &= 0 \end{aligned}$$

X and Y are not independent,

$\text{cov}(X,Y) = 0$ nonetheless

Propositions:

i. $\text{cov}(X,Y) = \text{cov}(Y,X)$

ii. $\text{cov}(X,X) = \text{var}(X)$

Proof: $\text{cov}(X,X) = E[(x-\mu_X)(x-\mu_X)]$
$= E[(x-\mu_X)^2]$
$= \text{var}(X)$

iii. $\text{cov}(aX,Y) = a\cdot\text{cov}(X,Y)$

Proof: $E[(ax-a\mu_X)(y-\mu_Y)]$
$= aE[(x-\mu_X)(y-\mu_Y)]$

iv. $\text{cov}(X+Y,Z) = \text{cov}(X,Z) + \text{cov}(Y,Z)$

Proof: $\text{cov}(X+Y,Z)$
$= E[(X+Y)Z] - E[X+Y]E[Z]$
$= E[XZ+YZ] - (E[X]+E[Y])E[Z]$
$= E[XZ] - E[X]E[Z] + E[YZ] - E[Y]E[Z]$
$= \text{cov}(X,Z) + \text{cov}(Y,Z)$

v. Generalization of the previous:

$$\text{cov}\left(\sum_{i=1}^{n} X_i, \sum_{j=1}^{m} Y_j\right) = \sum_{i=1}^{n}\sum_{j=1}^{m} \text{cov}(X_i, Y_j)$$

i.e. covariance is additive.

Proof: $\text{cov}(\Sigma X_i, \Sigma Y_i) = E[(\Sigma x_i - E[\Sigma x_i])(\Sigma Y_j - E[\Sigma Y_j])]$
$= E[(\Sigma X_i - \Sigma \mu_i)(\Sigma Y_j - \Sigma \nu_j)]$
$= E[\Sigma(X_i-\mu_i)\cdot\Sigma(Y_j-\nu_j)]$
$= E\left[\sum_i\sum_j (X_i-\mu_i)(Y_j-\nu_j)\right]$
$= \Sigma\Sigma\, E[(X_i-\mu_i)(Y_j-\nu_j)]$
$= \Sigma\Sigma\, \text{cov}(X_i,Y_i)$

$\mu_i \rightarrow$ mean of X_i

$\nu_i \rightarrow$ mean of Y_i

vi. $cov(X+c, Y) = cov(X,Y)$, c is a constant

Proof:
$$\begin{aligned} cov(X+c, Y) &= E[(X+c-\mu_{X+c})(Y-\mu_Y)] \\ &= E[(X+c-(\mu_X+c))(Y-\mu_Y)] \\ &= E[(X-\mu_X)(Y-\mu_Y)] = cov(X,Y) \end{aligned}$$

vii. $Var(X+Y) = Var(X) + Var(Y) + 2\,cov(X,Y)$

Proof:
$$\begin{aligned} var(X+Y) &= cov(X+Y, X+Y) \\ &= cov(X,X) + cov(X,Y) + cov(Y,X) + cov(Y,Y) \\ &= var(X) + 2\,cov(X,Y) + var(Y) \end{aligned}$$

viii. Generalization of the previous one:

$$Var\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} var(X_i) + 2\sum_{i<j}\sum cov(X_i, X_j)$$

proof:
$$\begin{aligned} &var\left(\sum_{i=1}^{n} X_i\right) \\ &= cov\left(\sum_{i=1}^{n} X_i, \sum_{j=1}^{n} X_j\right) \\ &= \sum_{i=1}^{n}\sum_{j=1}^{n} cov(X_i, X_j) \\ &= cov(X_1,X_1) + cov(X_1,X_2) + \cdots + cov(X_1,X_n) \\ &\quad cov(X_2,X_1) + cov(X_2,X_2) + \cdots + cov(X_2,X_n) \\ &\quad + \\ &\quad \vdots \\ &\quad + \\ &\quad cov(X_n,X_1) + \cdots + cov(X_n,X_n) \quad \leftarrow \sum var(X_i) \\ &= \sum var(X_i) + 2\sum_{i<j}\sum cov(X_i, X_j) \end{aligned}$$

ix. $Var(aX+bY+c) = a^2 \cdot var(X) + b^2 var(Y) + 2ab\,cov(X,Y)$

Suppose X and Y are discrete with joint *pmf* given by that of the basic example (*)

X \ Y	0	1	2	3	
0	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{1}{8}$	0	$\frac{1}{2}$
1	0	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{1}{8}$	$\frac{1}{2}$
	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$	

(i) Compute Cov(X, Y)

(ii) Compute $\rho_{X,Y}$ → correlation = $\frac{Cov}{\sigma_X \sigma_Y}$ ✓

→ $cov(X,Y) = E[XY] - E[X]E[Y]$

$= \left[1\cdot 1\cdot\frac{1}{8} + 1\cdot 2\cdot\frac{2}{8} + 1\cdot 3\cdot\frac{1}{8}\right] - \left[1\cdot\frac{1}{2}\right]\left[1\cdot\frac{3}{8} + 2\cdot\frac{3}{8} + 3\cdot\frac{1}{8}\right]$

$= \frac{1}{4}$

→ $\rho_{X,Y}$:

$E[X^2] = 1\cdot\frac{1}{2} = \frac{1}{2}$ $\quad E[Y^2] = 1^2\cdot\frac{3}{8} + 2^2\cdot\frac{3}{8} + 3^2\cdot\frac{1}{8} = 3$

$var[X] = E[X^2] - (E[X])^2$ $\quad var[Y] = 3 - \left(\frac{3}{2}\right)^2 = \frac{3}{4}$

$= \frac{1}{2} - \left(\frac{1}{2}\right)^2 = \frac{1}{4}$

$\sigma_X = \frac{1}{2}$ $\quad \sigma_Y = \frac{\sqrt{3}}{2}$

$$\rho_{X,Y} = \frac{cov(X,Y)}{\sigma_X \sigma_Y} = \frac{\frac{1}{4}}{\frac{1}{2}\cdot\frac{\sqrt{3}}{2}} = \frac{1}{\sqrt{3}}$$

Correlation of X and Y is normalizing the covariance. $\rho(X,Y)$
We define the standardized versions of X and Y as

$$u = \frac{X - E[X]}{\sigma_X} \qquad v = \frac{Y - E[X]}{\sigma_Y}$$

Then, correlation $(X,Y) = \rho(X,Y) = cov(u,v)$

$$= cov\left(\frac{X - E[X]}{\sigma_X}, \frac{Y - E[Y]}{\sigma_Y}\right)$$

$$= cov\left(\frac{X}{\sigma_X}, \frac{Y}{\sigma_Y}\right)$$

$$= \frac{cov(X,Y)}{\sigma_X \sigma_Y}$$

7.20. *The Conditional Covariance Formula.* The conditional covariance of X and Y, given Z, is defined by

$$\mathrm{Cov}(X, Y|Z) = E[(X - E[X|Z])(Y - E[Y|Z])|Z]$$

(a) Show that

$$\mathrm{Cov}(X, Y|Z) = E[XY|Z] - E[X|Z]E[Y|Z]$$

(b) Prove the conditional covariance formula

$$\begin{aligned}\mathrm{Cov}(X, Y) = {} & E[\mathrm{Cov}(X, Y|Z)] \\ & + \mathrm{Cov}(E[X|Z], E[Y|Z])\end{aligned}$$

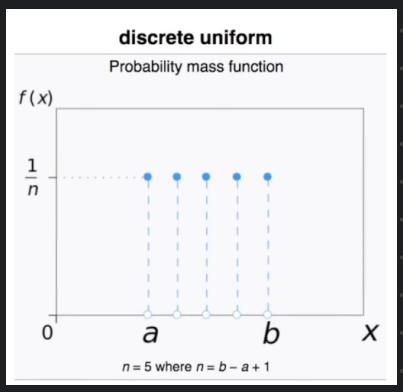
(c) Set $X = Y$ in part (b) and obtain the conditional variance formula.

Discrete Uniform Random Variable:

A uniform random variable is one with equal probabilities for all the entire sample space.

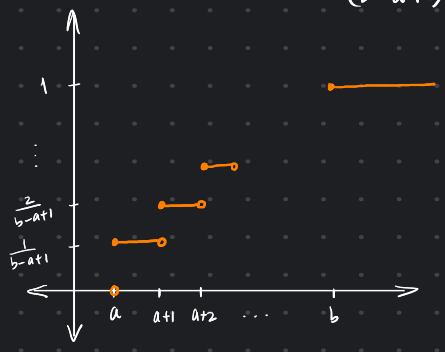
The shorthand X ~ discrete uniform(a, b) is used to indicate that the random variable X has the discrete uniform distribution with integer parameters a and b, where a < b.
A discrete rv X with parameters a and b has the PMF:

$$f(x) = \frac{1}{b-a+1}, \quad x = a, a+1, \cdots, b$$



e.g. we roll a die. X is the # appearing on the top. X ~ discrete uniform(1, 6)

CDF: $F(x) = P(X \le x) = \frac{(x-a+1)}{(b-a+1)}$



Expectation:

PMF	x	a	$a+1$	$\cdots$	b
	$P(X=x)$	$\frac{1}{n}$	$\frac{1}{n}$	$\cdots$	$\frac{1}{n}$

$$E[X] = \sum x_i \cdot P(X = x_i)$$

$$= \frac{1}{n}(a + a+1 + \cdots + b) = \frac{1}{n} \cdot \frac{n}{2}(a+b) = \frac{a+b}{2}$$

$$E[X] = \frac{a+b}{2}$$

$$\text{Var}[X] = E[X^2] - (E[X])^2$$

$$= \frac{(b-a+1)^2 - 1}{12}$$

eg: 10 ping pong balls are numbered 1–10 and placed in a bag.
One ball is selected randomly from the bag.

(i) Find the expected number on the ball.

x: 1 2 3 ... 10

$\rightarrow E[X] = \frac{a+b}{2} = \frac{1+10}{2} = 5.5$

(ii) Find the variance of number on the ball.

$\rightarrow Var[X] = \frac{(b-a+1)^2-1}{12} = \frac{10^2-1}{12} = 8.25$

$10^2-1 = 99$

(iii) Find the probability that the number drawn is between 7 and 10 (both are not inclusive)

Sol: $P(7 < x < 10) = P(x=8) + P(x=9)$

$= 1/10 + 1/10 = \boxed{1/5}$

Bernoulli Random Variable

Consider a trial, whose outcome can be a failure(X=0) or a success(X=1).
The PMF of X can be given as

$X : \{0, 1\}$

$P(0) = P\{X=0\} = 1-p = q$
$P(1) = P\{X=1\} = p$

where p, 0 <= p <= 1, is the probability that the trial is a success.

e.g. You toss a coin, the outcome is either heads or tails.
ii. A child is born. The gender of the child is either male or female.

A rv X is a Bernoulli rv with parameter p: X ~ Bernoulli (p)
if its PMF is given as

$$P_X(x) = \begin{cases} p, & x=1 \\ 1-p, & x=0 \\ 0, & \text{otherwise} \end{cases}$$

An indicator random variable is a Bernoulli rv.

CDF:

$$F(x) = \begin{cases} 0 & \text{if } x<0 \\ 1-p & \text{if } 0 \le x < 1 \\ 1 & \text{if } x \ge 1 \end{cases}$$

$E[X] = 0(1-p) + 1(p) = p$

$E[X^2] = 0^2(1-p) + 1^2(p) = p$

$Var(X) = p - p^2 = p(1-p) = pq$

$Var(X) \in [0, 1/4]$ at $p = \frac{1}{2}$

Binomial random variable

Suppose that n independent trials, each with the probability of success p and that of failure 1-p are to be performed.
If X represents the # of successes that occur in n trials, then X is said to be a binomial rv with parameters (n, p).

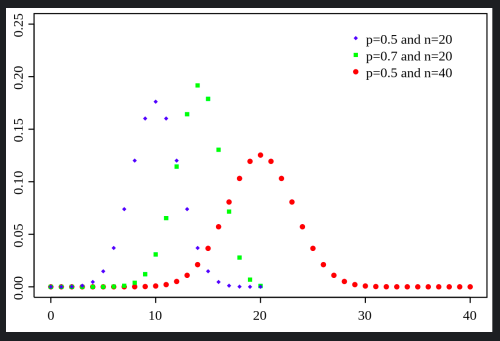
The PMF of a binomial rv with parameters (n, p) is $P\{X=i\} = \binom{n}{i} p^i (1-p)^{n-i}$

Sum of probabilities $= \sum_{i=0}^{n} \binom{n}{i} p^i (1-p)^{n-i} = (p+(1-p))^n = 1$

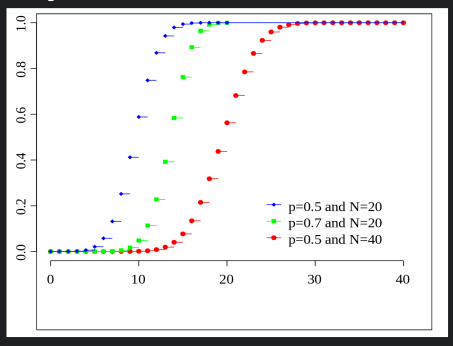
A rv X is a Binomial rv with parameters (n, p),
X ~ B(n, p),
if its PMF is

$$P_X(k) = \begin{cases} {}^nC_k\, p^k (1-p)^{n-k}, & k = 0, 1, 2, \ldots, n \\ 0, & \text{otherwise} \end{cases}$$

PMF:



CDF:



$$E[X] = \sum_{n=0}^{n} \left({}^nC_n\, p^n q^{n-n} \right) \cdot n$$

or you could think of a Binomial variable as the sum of n Bernoulli variables.

$X = X_1 + X_2 + \ldots + X_n$

X_1: 1 (p), 0 ($1-p$, q)

$E[X] = E[X_1] + E[X_2] + \ldots + E[X_n]$

$= p + p + \ldots + p$

$= np$

$E[X] = np$

If $X_i \sim \text{Bernoulli}(p)$ $\forall i = 1, 2, \ldots, n$, then $X = X_1 + X_2 + \ldots + X_n = \sum_{i=1}^{n} X_i$, $X \sim B(n, p)$

$\text{Var}[X] = \text{Var}[X_1] + \text{Var}[X_2] + \ldots + \text{Var}[X_n]$

$= pq + pq + \ldots + pq$

... as all n binomial trials are independent of one another.

Eg: It is known that screws produced by a certain company will be defective with probability 0.01 independent of each other. The company sells the screws in package of 10 and offers a money back guarantee that at most 1 out of 10 screws is defective. What proportion of packages sold will the company replace?

→ $p = 0.01$ (defective) $\quad n = 10 \quad P(x \geq 2) = ? \quad X$: # of defective screws.

$$P(X \geq 2) = 1 - P(X < 2)$$
$$= 1 - P(X=1) - P(X=0)$$
$$= 1 - \left[{}^{10}C_1 (0.01)^1 (1-0.01)^9\right] - \left[{}^{10}C_0 (0.01)^0 (1-0.01)^{10}\right]$$
$$= 1 - (10)(0.01)(0.99)^9 - (1)(1)(0.99)^{10}$$
$$= 0.0042$$
$$= 0.42\% \text{ of all packages sold.}$$

EXAMPLE 5.1b The color of one's eyes is determined by a single pair of genes, with the gene for brown eyes being dominant over the one for blue eyes. This means that an individual having two blue-eyed genes will have blue eyes, while one having either two brown-eyed genes or one brown-eyed and one blue-eyed gene will have brown eyes. When two people mate, the resulting offspring receives one randomly chosen gene from each of its parents' gene pair. If the eldest child of a pair of brown-eyed parents has blue eyes, what is the probability that exactly two of the four other children (none of whom is a twin) of this couple also have blue eyes?

→ Parents are brown-eyed, but eldest child is blue eyed.
so there must be 1 blue gene in both as for blue eyes we need bl + bl.
(bl: from parent 1, bl: from parent 2)

Parents are Br Bl + Br Bl

Pr(exactly 2 of 4 other children have blue eyes) = ?

P(child of that couple has blue eyes) = $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$.

X: # of blue-eyed children $\quad X \sim \text{Binomial}(4, \frac{1}{4})$ (n, p)

$$P(X=2) = {}^nC_2\, p^2\, q^{n-2}$$
$$= {}^4C_2 \cdot \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{3}{4} \cdot \frac{3}{4}$$
$$= \frac{4 \cdot 3}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{3}{4} \cdot \frac{3}{4}$$
$$= \frac{27}{2^7} = \frac{27}{128} = 0.2109$$

EXAMPLE 5.1c A communications system consists of n components, each of which will, independently, function with probability p. The total system will be able to operate effectively if at least one-half of its components function.

(a) For what values of p is a 5-component system more likely to operate effectively than a 3-component system?

(b) In general, when is a $2k+1$ component system better than a $2k-1$ component system?

→ X: # of functioning components.

If $X \geq n/2$, the system operates effectively.

$P(X \geq n/2)$ → prob. that the system works.

For $n=5$, $P(X \geq 3) = P(X=3) + P(X=4) + P(X=5)$

$$= {}^5C_3\, p^3(1-p)^2 + {}^5C_4\, p^4(1-p)^1 + {}^5C_5\, p^5(1-p)^0$$

$$= 10p^3(1-p)^2 + 5p^4(1-p) + p^5$$

For $n=3$, $P(X \geq 2) = {}^3C_2\, p^2(1-p)^1 + {}^3C_3\, p^3(1-p)^0$

$$= 3p^2(1-p) + p^3$$

$$10p^3(1-p)^2 + 5p^4(1-p) + p^5 \geq 3p^2(1-p) + p^3$$

$$10p(1-p)^2 + 5p^2(1-p) + p^3 \geq 3(1-p) + p$$

$$10p(p^2 - 2p + 1) + 5p^2 - 4p^3 \geq 3 - 2p$$

$$6p^3 - 15p^2 + 12p - 3 \geq 0$$

$$\Rightarrow p \geq \frac{1}{2}$$

Properties of binomial rvs:

- If X(m, p) and Y(n, p) are binomial rvs, then Z = X+Y is a Bernoulli rv with parameters (m+n, p).

$$Z = X + Y$$
$$= \underbrace{X_1 + X_2 + \dots + X_m}_{\text{m-independent Bernoulli trials}} + \underbrace{Y_1 + \dots + Y_n}_{\text{n-independent Bernoulli trials}}$$

$\underbrace{\qquad\qquad}_{\text{m+n independent Bernoulli trials}}$

- If X ~ B(n, p), then as k goes from 0 to n, P(X=k) first increases and then decreases (like a bell curve), reaching its maximum value when k is the largest integer <= (n+1)p.

$$\frac{P(X=k)}{P(X=k-1)} = \frac{\binom{n}{k} p^k (1-p)^{n-k}}{\binom{n}{k-1} p^{k-1} (1-p)^{n-k+1}} = \frac{\frac{n!}{k!(n-k)!} \cdot p}{\frac{n!}{(k-1)!(n-k+1)!} \cdot (1-p)} = \frac{(n-k+1) \cdot p}{k \cdot (1-p)}$$

we'll check till what point $P(X=k) \geq P(X=k-1)$
i.e. $p \uparrow$ if $k \uparrow$

$\therefore P(X=k) \geq P(X=k-1)$ iff

$$(n-k+1)p \geq k(1-p)$$
$$np - kp + p \geq k - kp$$
$$k \leq p(n+1)$$

⟶ so $P(X=k)$ keeps increasing till $k = p(n+1)$, then starts decreasing.

$\therefore P(X=k)$ is maximum at $k = \lfloor p(n+1) \rfloor$

$$P(X \leq i) = P(X=0) + P(X=1) + \dots + P(X=i)$$
$$= \sum_{k=0}^{i} \binom{n}{k} p^k (1-p)^{n-k}$$

$$P(X=k+1) = \frac{p}{1-p} \cdot \frac{n-k}{k+1} P(X=k)$$

← We can calculate the prob for k+1 if we know that for k

This is done in computer programs, since calculating nCk is computationally expensive.

e.g. X(6, 0.4)

$P(X=0) = {}^6C_0 (0.4)^0 (0.6)^6 = 0.0467$

$P(X=1) = \frac{0.4}{0.6} \times \frac{6}{1} \times 0.0467 = 0.1866$

$P(X=2) = \frac{0.4}{0.6} \times \frac{5}{2} \times 0.1866 = 0.311$

Poisson random variable

It is a limiting case of the binomial rv.

A rv X, taking the values 0, 1, 2, 3, ... is a Poisson rv with parameter λ, λ>0, if its PMF is given by

$$P(x=n) = \frac{e^{-\lambda}\lambda^{x}}{x!}, \qquad x = 1, 2, 3, \ldots, \infty$$

Sum of probabilities:

$$\sum_{n=0}^{\infty} \frac{e^{-\lambda}\lambda^{n}}{n!} = e^{-\lambda}\sum_{n=0}^{\infty}\frac{\lambda^{n}}{n!}$$

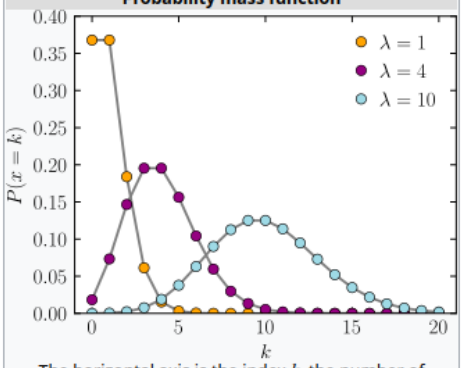
$$= e^{-\lambda}e^{\lambda}$$

$$= 1$$

$$e^{a} = \sum_{i=0}^{\infty}\frac{a^{i}}{i!}$$

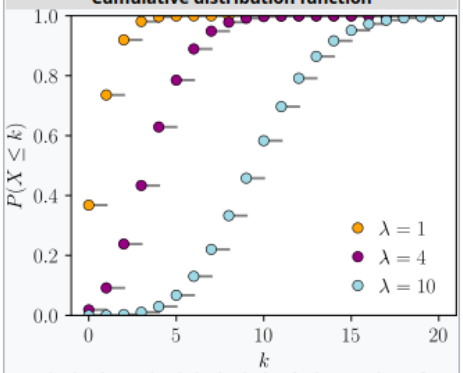
Poisson Distribution

Probability mass function



The horizontal axis is the index k, the number of occurrences. λ is the expected rate of occurrences. The vertical axis is the probability of k occurrences given λ. The function is defined only at integer values of k; the connecting lines are only guides for the eye.

Cumulative distribution function



The horizontal axis is the index k, the number of occurrences. The CDF is discontinuous at the integers of k and flat everywhere else because a variable that is Poisson distributed takes on only integer values.

Notation	$\mathrm{Pois}(\lambda)$
Parameters	$\lambda \in (0, \infty)$ (rate)
Support	$k \in \mathbb{N}$ (Natural numbers starting from 0)
PMF	$\frac{\lambda^k e^{-\lambda}}{k!}$
CDF	$\frac{\Gamma(\lfloor k+1 \rfloor, \lambda)}{\lfloor k \rfloor !}$, or $e^{-\lambda}\sum_{j=0}^{\lfloor k \rfloor}\frac{\lambda^j}{j!}$, or $Q(\lfloor k+1 \rfloor, \lambda)$ (for $k \geq 0$, where $\Gamma(x, y)$ is the upper incomplete gamma function, $\lfloor k \rfloor$ is the floor function, and Q is the regularized gamma function)
Mean	λ
Median	$\approx \left\lfloor \lambda + \frac{1}{3} - \frac{1}{50\lambda} \right\rfloor$
Mode	$\lceil \lambda \rceil - 1, \lfloor \lambda \rfloor$
Variance	λ
Skewness	$\frac{1}{\sqrt{\lambda}}$
Excess kurtosis	$\frac{1}{\lambda}$
Entropy	$\lambda\left[1 - \log(\lambda)\right] + e^{-\lambda}\sum_{k=0}^{\infty}\frac{\lambda^k \log(k!)}{k!}$ or for large λ $\approx \frac{1}{2}\log(2\pi e\lambda) - \frac{1}{12\lambda} - \frac{1}{24\lambda^2} - \frac{19}{360\lambda^3} + \mathcal{O}\left(\frac{1}{\lambda^4}\right)$
MGF	$\exp\left[\lambda\left(e^t - 1\right)\right]$
CF	$\exp\left[\lambda\left(e^{it} - 1\right)\right]$
PGF	$\exp\left[\lambda\left(z - 1\right)\right]$
Fisher information	$\frac{1}{\lambda}$

Mean and variance of Poisson rv

$X \sim \text{Pois}(\lambda)$

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}, \qquad x = 0, 1, 2, \ldots, \infty$$

$$E[X] = \sum_{x=0}^{\infty} x \cdot P(X = x)$$

$$= \sum_{x=0}^{\infty} x \frac{e^{-\lambda} \lambda^x}{x!}$$

$$= 0 + \sum_{x=1}^{\infty} x \frac{e^{-\lambda} \lambda^x}{x!}$$

$$= \sum_{x=1}^{\infty} x e^{-\lambda} \frac{\lambda^x}{x!}$$

$$= e^{-\lambda} \lambda \sum_{x=1}^{\infty} \frac{\lambda^{x-1}}{(x-1)!}$$

$$= e^{-\lambda} \lambda \sum_{x=0}^{\infty} \frac{\lambda^x}{x!}$$

$$= e^{-\lambda} \lambda e^{\lambda}$$

$$= \lambda$$

$$E[X^2] = \sum_{x=0}^{\infty} x^2 \cdot P(X = x)$$

$$=$$

$$\text{Var}[X] = E[X^2] - (E[X])^2$$

Sum of 2 independent Poisson rvs is also a Poisson rv.

$X_1 \sim \text{Pois}(\lambda_1)$

$X_2 \sim \text{Pois}(\lambda_2)$

$$X_1 + X_2 \sim \text{Pois}(\lambda_1 + \lambda_2)$$

Poisson rv as an approximation of binomial rv:

Poisson rv is an approximation of binomial rv with parameters (n, p) if n is large and p is small enough s.t. np is moderate.

$$\text{Poisson}(\lambda) \approx \lim_{\substack{n\to\infty \\ p\to 0}} \text{Binomial}(n, p)$$

$X \sim B(n, p)$

$$P(X=x) = \binom{n}{x} p^x (1-p)^{n-x}$$

$$= \frac{n!}{(n-x)!\,x!} \cdot p^x \cdot (1-p)^{n-x}$$

$$= \frac{n!}{(n-x)!\,x!} \left(\frac{\lambda}{n}\right)^x \cdot \left(1-\frac{\lambda}{n}\right)^{n-x} \qquad \text{let } \lambda = np$$

$$= \frac{n(n-1)\cdots(n-(x-1))(n-x)!}{(n-x)!\,x!} \cdot \left(\frac{\lambda}{n}\right)^x \left(1-\frac{\lambda}{n}\right)^{n-x}$$

$$= \frac{n(n-1)\cdots(n-x+1)}{n^x} \cdot \frac{\lambda^x}{x!} \cdot \frac{(1-\lambda/n)^n}{(1-\lambda/n)^x}$$

$$= 1 \cdot \frac{\lambda^x}{x!} \cdot \frac{e^{-\lambda}}{1}$$

$$P(X=x) = \frac{e^{-\lambda}\lambda^x}{x!}$$

$$\lim_{n\to\infty} \frac{n(n-1)\cdots(n-x+1)}{n^x} = 1$$

$$\lim_{n\to\infty} \left(1-\frac{\lambda}{n}\right)^n = e^{\lim_{n\to\infty}(1-\frac{\lambda}{n}-1)n} = e^{-\lambda}$$

$$\lim_{n\to\infty} \left(1-\frac{\lambda}{n}\right)^x = 1$$

eg. the # of misprints in a book,
the # of wrong telephone #s dialled in a day,
the # of accidents that happen in a day.

} In all these, n is very high, p is very low.

Q. Suppose that the # of typos on a single page has a Poisson distribution ($\lambda = 1/2$). Calculate the prob. that there's atleast one error on any page.

→ $\lambda = 1/2 = np$

$$P(X=x) = \frac{e^{-\lambda}\lambda^x}{x!}$$

$$P(X \geq 1) = 1 - P(X=0)$$

$$= 1 - \frac{e^{-\lambda}\lambda^0}{0!}$$

$$= 1 - e^{-1/2} = 0.3934$$

Q. Avg # of accidents per week on a highway = 3. Calculate the probability that there are less than 3 accidents this week.

→ Assume that # of accidents $\sim$ Pois(λ)

$\lambda = 3 = np$

$$P(X \leq 3) = P(X=0) + P(X=1) + P(X=2)$$

$$= \sum_{x=0}^{3} \frac{e^{-\lambda}\lambda^x}{x!} = e^{-3}\left[\frac{\lambda^0}{0!} + \frac{\lambda^1}{1!} + \frac{\lambda^2}{2!}\right] = e^{-3}\left[1 + 3 + \frac{9}{2}\right] = e^{-3} \cdot \frac{17}{2} = 0.4231$$

Q: It is known that screws produced by a certain company will be defective with probability 0.01 independent of each other. The company sells the screws in package of 10 and offers a money back guarantee that at most 1 out of 10 screws is defective. What proportion of packages sold will the company replace?

same question done previously using Binomial distribution, ans = 0.42%

→ In 10 packages, # of defective screws = $np = 10 \cdot (0.01) = 0.1$

$\lambda = 0.1$ $\quad X \sim Pois(\lambda = 0.1)$

$P(X > 1) = 1 - (P[X=0] + P[X=1])$

$= 1 - e^{-0.1}\left[\frac{\lambda^0}{0!} + \frac{\lambda^1}{1!}\right] = 1 - e^{-0.1}\left[1 + 0.1\right] = 1 - e^{-0.1}(1.1) = 0.004678 = 0.4678\ \%$

Q. If the avg. # of claims handled daily by an insurance company is 5.

a. What proportion of days have less than 3 claims?

→ $\lambda = 5$

$P(X<3) = P(X=0) + P(X=1) + P(X=2)$

$= e^{-5}\left[1 + 5 + \frac{25}{2}\right]$

$= e^{-5} \cdot \frac{37}{2} = 0.1246 \approx 12.46\ \%$

b. What is the probability that there'll be exactly 4 claims in 3 of the next 5 days? Assume that the # of claims on different days are independent.

→ $P(4 \text{ claims}) = P(X=4) = e^{-5}\frac{5^4}{4!} = \frac{625}{24} \cdot e^{-5} = 0.1754$

$p = 0.1754, \quad q = 1 - p = 0.8245$

Now use $Binom(5, 0.1754)$ (n, p)

of claims in a day follows $Pois(\lambda = 5)$

The later part is assumed to follow $B(n=5, p = \text{prob of getting 4 claims})$

$P(X=3) = {}^5C_3 \cdot (0.1754)^3 (0.8245)^2$

$= 10 \cdot (0.0054)(0.6798)$

$= 0.0367$

Poisson paradigm

Consider n events, with p_i equal to the probability that event i occurs, i = 1, 2, ..., n.
If all p_i are small, and the trials are independent or even weakly dependent, then the # of these events that occur approximately has a Poisson distribution with mean $\sum_{i=1}^{n} p_i$.

E.g. At a party, n people put their hats in the center of a room, where they are mixed up. Each person then randomly chooses a hat. Let X denote the # of people who select their own hat. For a large n, X ~ Pois(1).

→ $X: \{0, 1, 2, \dots, n\}$

Let X_i : i^{th} person takes their own hat. [indicator $\begin{matrix}1\\0\end{matrix}$]

$X_1, X_2, \dots, X_n$ are not independent. Weakly dependent is fine as for large n, $\frac{1}{n} \approx \frac{1}{n-1}$

$X = \sum_{i=1}^{n} X_i$

$E[X] = E[X_1] + E[X_2] + \dots + E[X_n]$ $\qquad E[X_i] = 1 \cdot P(i \text{ takes own hat}) = \frac{1}{n}$

$= \frac{1}{n} \cdot n = 1$

$E[X] = 1$

$X \sim Pois(\lambda = 1)$

Q. The # of defective stereos produced daily at a certain plant follows a Poisson distribution with mean 4. Over a 2-day span, what is the probability that the # of defective stereos doesn't exceed 3?

→ X_1: day 1 defective stereos $\quad X_2$: day 2

$X_1 \sim \text{Pois}(\lambda = 4)$ $\quad X_2 \sim \text{Pois}(\lambda = 4)$

$X = X_1 + X_2$: # of defective stereos in 2 days.

$X \sim \text{Pois}(\lambda = 8)$

$$P(X \le 3) = e^{-8}\left[\frac{8^0}{0!} + \frac{8^1}{1!} + \frac{8^2}{2!} + \frac{8^3}{3!}\right] = 0.04238$$

Given a mean for t time duration in a Poisson distribution, and we want to find the probability for k time duration, we can consider it a Poisson distribution.

$\lambda_t \rightarrow$ mean for duration t

$\frac{\lambda_t}{t} \rightarrow$ for 1 unit

k units $\rightarrow \frac{\lambda_t}{t} \cdot k = \lambda_k$

↳ $\text{Pois}(\lambda = \frac{\lambda_t}{t} \cdot k)$

Q. The # of emails that I get in a weekday can be modeled as a Poisson distribution with an average of 0.2 emails per minute.

a. What is the probability that I get no emails in an interval of length 5 minutes?

→ # of emails in 1 min $\sim \text{Pois}(\lambda = 0.2)$

⇒ # of emails in 5 mins $\sim \text{Pois}(\lambda = \frac{0.2}{1} \times 5)$ i.e. $\text{Pois}(\lambda = 1)$

$$P(X=0) = e^{-1}\frac{(-1)^0}{0!} = \frac{1}{e} \approx 0.3679$$

b. What is the probability that I get more than 3 emails in an interval of 10 minutes?

→ # of emails in 10 mins $\sim \text{Pois}(\lambda = \frac{0.2}{1} \times 10)$ i.e $\text{Pois}(\lambda = 2)$

$$P(X>3) = 1 - e^{-2}\left[\frac{2^0}{0!} + \frac{2^1}{1!} + \frac{2^2}{2!} + \frac{2^3}{3!}\right]$$

$$= 1 - e^{-2}\left[1 + 2 + 2 + \frac{4}{3}\right] = 1 - e^{-2}\cdot\frac{19}{3} = 0.1428$$

Calculating Poisson distribution in a computer program

$X \sim \text{Pois}(\lambda)$

$$\frac{P(X=i+1)}{P(X=i)} = \frac{e^{-\lambda}\lambda^{i+1}}{(i+1)!}\cdot\frac{i!}{e^{-\lambda}\lambda^i} = \frac{\lambda}{i+1}$$

$$P(X=i+1) = \frac{\lambda}{i+1}\cdot P(X=i)$$

e.g. $P(X=2) = \frac{\lambda}{2}\cdot P(X=1)$

GATE CSE 2017 Set 2 | Question: 48

If a random variable X has a Poisson distribution with mean 5, then the expectation $E\left[(x+2)^2\right]$ equals ___.

→ $\lambda = 5$

$$E[(x+2)^2] = E[x^2] + 4E[x] + 4$$
$$= [\text{Var}(x) + (E(x))^2] + 4E[x] + 4$$
$$= 5 + 25 + 4(5) + 4$$
$$= 54$$

GATE CSE 2013 | Question: 2

Suppose p is the number of cars per minute passing through a certain road junction between 5 PM and 6 PM, and p has a Poisson distribution with mean 3. What is the probability of observing fewer than 3 cars during any given minute in this interval?

A. $\frac{8}{(2e^3)}$
B. $\frac{9}{(2e^3)}$
C. $\frac{17}{(2e^3)}$ ✓
D. $\frac{26}{(2e^3)}$

→ $P \sim \text{Pois}(\lambda = 3)$

$$P(p<3) = e^{-3}\left[\frac{3^0}{0!} + \frac{3^1}{1!} + \frac{3^2}{2!}\right]$$
$$= e^{-3}\left[1 + 3 + \frac{9}{2}\right]$$
$$= \frac{17}{2e^3}$$

An observer counts 240 veh/h at a specific highway location. Assume that the vehicle arrival at the location is Poisson distributed, the probability of having one having vehicle arriving over a 30-second time interval is _____

→ 30-sec interval ~ Pois $\left(\lambda = \frac{240}{60 \cdot 60} \cdot 30 = 2\right)$

Pois$(\lambda = 2)$

$$P(x=1) = \frac{e^{-2}\, 2^1}{1!} = \frac{2}{e^2} = 0.2706$$

Geometric rv

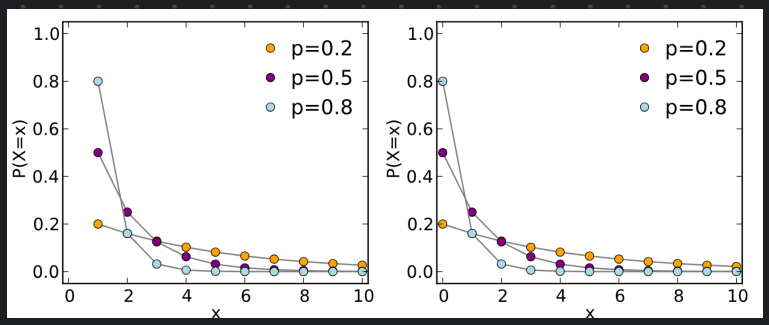
Consider a Bernoulli trial process done infinite times. If X is a geometric rv, P(X=a) is the probability that the first success is obtained in the ath trial.

$$P(X=k) = (1-p)^{k-1} p \qquad k = 1, 2, 3, \ldots$$

The geometric distribution gives the probability that the first occurrence of success requires k independent trials, each with success probability p. If the probability of success on each trial is p, then the probability that the k-th trial is the first success is

$$\Pr(X = k) = (1-p)^{k-1} p$$

for $k = 1, 2, 3, 4, \ldots$



Sum of probabilities :

$$\sum_{k=1}^{\infty} (1-p)^{k-1} p = p + p(1-p) + p(1-p)^2 + \cdots$$

$$= \frac{p}{1-(1-p)} = 1 \qquad (1-p) < 1$$

Expected value :

$$E[X] = \sum_{i=1}^{\infty} i \cdot (1-p)^{i-1} p$$

$$= p \sum_{i=1}^{\infty} i (1-p)^{i-1}$$

$$= p \frac{1}{p^2}$$

$$E[X] = \frac{1}{p}$$

$$\sum_{i=1}^{\infty} i a^{i-1} = 1a^0 + 2a^1 + 3a^2 + \cdots$$

$$S = 1a^0 + 2a^1 + 3a^2 + \cdots$$

$$Sa = \quad 1a^1 + 2a^2 + \cdots$$

$$S(1-a) = 1a^0 + 1a^1 + 1a^2 + \cdots = \frac{1}{1-a}$$

$$S = \frac{1}{(1-a)^2} = \frac{1}{p^2}$$

or $$E[X] = \sum_{i=1}^{\infty} (i-1+1)(1-p)^{i-1} p$$

$$= \sum_{i=1}^{\infty} (i-1)(1-p)^{i-1} p + \sum_{i=1}^{\infty} 1 \cdot (1-p)^{i-1} p$$

$$= \sum_{j=0}^{\infty} j (1-p)^j p + \sum_{i=1}^{\infty} (1-p)^{i-1} p$$

$$= (1-p) \sum_{j=1}^{\infty} j (1-p)^{j-1} p + 1$$

$$E[X] = (1-p) E[X] + 1$$

$$\Rightarrow E[X] = \frac{1}{p}$$

$$E[X^2] = \sum_{i=1}^{\infty} (i+1-1)^2 (1-p)^{i-1} p$$

Q. An urn contains N white and M black balls. Balls are randomly selected, one at a time, with replacement until a black ball is obtained. What is the probability that

a. exactly n draws are needed?

→ $p = \frac{M}{M+N}$ $(1-p) = \frac{N}{M+N}$

It is done with replacement so it is a geometric distribⁿ.

$X \sim \text{Geom}\left(p = \frac{M}{M+N}\right)$

$$P(X=n) = \left(\frac{N}{M+N}\right)^{n-1}\left(\frac{M}{M+N}\right)$$

b. at least k draws are needed?

→ $P(X=k) + P(X=k+1) + \cdots$

$$= \left(\frac{M}{M+N}\right)\left(\frac{N}{M+N}\right)^{k-1} + \left(\frac{M}{M+N}\right)\left(\frac{N}{M+N}\right)^{k} + \left(\frac{M}{M+N}\right)\left(\frac{N}{M+N}\right)^{k+1} + \cdots$$

$$= \frac{\left(\frac{M}{M+N}\right)\cdot\left(\frac{N}{M+N}\right)^{k-1}}{1-\left(\frac{N}{M+N}\right)}$$

$$= \left(\frac{N}{M+N}\right)^{k-1}$$

OR we calc. P(failure) in k-1 attempts.
So at least k draws would be needed.

$P = (1-p)^{k-1}$

$= \left(\frac{N}{M+N}\right)^{k-1}$

Negative binomial rv

In n independent trials, each with a probability of success p, r successes are accumulated.
The negative binomial rv denotes the total # of trials required.

$X \sim NB(r, p)$

$$P(X = n) = \binom{n-1}{r-1} p^r (1-p)^{n-r}$$

n trials
r successes

$\underbrace{\times \times \times \cdots \times}_{n-1 \text{ trials}} \underbrace{\checkmark}_{n^{th} \text{ trial}}$
$(r-1)$ successes — r^{th} success

↳ choose which of these would be successes: $\binom{n-1}{r-1}$ · p^{r} (r successes) · $(1-p)^{n-r}$ ((n-r) failures)

> Different texts (and even different parts of this article) adopt slightly different definitions for the negative binomial distribution. They can be distinguished by whether the support starts at $k = 0$ or at $k = r$, whether p denotes the probability of a success or of a failure, and whether r represents success or failure,[1] so identifying the specific parametrization used is crucial in any given text.

Sum of Probabilities

$X \sim NB(r, p)$

$X = X_1 + X_2 + \cdots + X_r = \sum_{i=1}^{r} X_i$

$X_i \sim Geom(p)$ ← we want r successes. X_i would tell the # of trials for i^{th} success.
$i = 1, 2, \cdots, r$

$\sum X_i$ = (# of trials for 1st success) + (# of trials for 2nd success) + ⋯ + (# of trials for r^{th} success)

↳ Sum

$P(X = r) + P(X = r+1) + \cdots$

$= (P_{X_1=1} + P_{X_1=2} + \cdots)(P_{X_2=1} + P_{X_2=2} + \cdots) \cdots (P_{X_r=1} + P_{X_r=2} + \cdots)$

$= 1 \cdot 1 \cdot 1 \cdots 1$

$= 1$

> Suppose that a sequence of independent Bernoulli trials is performed, with $p = P(\text{"success"})$ for each trial. Fix an integer r to be greater than or equal to 2 and define the random variable X to give the number of trial at which the rth success occurs. Then X has a ***negative binomial distribution*** with parameters r and p. The probability mass function of X is given by
>
> $$\begin{aligned} p(x) = P(X = x) &= P(r^{th} \text{ success is on } x^{th} \text{ trial}) \\ &= \underbrace{P(1^{st}\ (r-1) \text{ successes in } 1^{st}\ (x-1) \text{ trials})}_{\text{bniomial with } n=x-1} \times P(r^{th} \text{ success on } x^{th} \text{ trial}) \\ &= \binom{x-1}{r-1} p^{r-1} (1-p)^{(x-1)-(r-1)} \times p \\ &= \binom{x-1}{r-1} p^{r} (1-p)^{x-r}, \quad \text{for } x = r, r+1, r+2, \ldots \end{aligned}$$

e.g. Throw a die. Success = getting a 1. $p = \frac{1}{6}$. Keep throwing die to get 3 successes.

X: # of trials reqd.

$X \sim NB(r=3, p=1/6)$

$X = X_1 + X_2 + X_3$ $\quad X_1 \sim Geom(1/6)$ # of trials to get 1st success

eg: Throw a die. success = getting a '1' $p(\text{success}) = 1/6$

keep throwing the die until you get 3 successes

$X \rightarrow$ # trials required to get 3 successes

$X \sim$ Neg. Binomial $(r=3, p=1/6)$ $\quad X \rightarrow \{3, 4, 5, 6, \ldots \infty\}$

$X = X_1 + X_2 + X_3$

$\downarrow$

$X_1 \sim$ Geometric $(1/6) \rightarrow$ # trials to get 1 success

$\downarrow$

$X_1 \sim$ Geometric $(1/6) \rightarrow$ # trials to get 1 success

$X = 3, \quad X_1 = 1, \; X_2 = 1, \; X_3 = 1$

$X = 4,$

X_1	X_2	X_3	
1	1	2	✓
1	2	1	✓
2	1	1	✓

Alternative definition for NB rv

In probability theory and statistics, the **negative binomial distribution** is a discrete probability distribution that models the number of failures in a sequence of independent and identically distributed Bernoulli trials before a specified (non-random) number of successes (denoted r) occurs.[2] For example, we can define rolling a 6 on a dice as a success, and rolling any other number as a failure, and ask how many failure rolls will occur before we see the third success ($r = 3$). In such a case, the probability distribution of the number of failures that appear will be a negative binomial distribution.

Imagine a sequence of independent Bernoulli trials: each trial has two potential outcomes called "success" and "failure." In each trial the probability of success is p and of failure is $1 - p$. We observe this sequence until a predefined number r of successes occurs. Then the random number of observed failures, X, follows the **negative binomial** (or **Pascal**) distribution:

$$X \sim \mathrm{NB}(r, p)$$

The **Pascal distribution** (after Blaise Pascal) and **Polya distribution** (for George Pólya) are special cases of the negative binomial distribution. A convention among engineers, climatologists, and others is to use "negative binomial" or "Pascal" for the case of an integer-valued stopping-time parameter (r) and use "Polya" for the real-valued case.

$$f(k; r, p) \equiv \Pr(X = k) = \binom{k + r - 1}{k} (1 - p)^k p^r$$

where *r* is the number of successes, *k* is the number of failures, and *p* is the probability of success on each trial.

Negative Binomial Random variable

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$$P\{X=x\} = \binom{x-1}{r-1} p^r (1-p)^{x-r} \qquad x = r, r+1, r+2, \cdots \infty$$

$X \sim$ Neg. Binomial (r, p)

$X = X_1 + X_2 + X_3 + \cdots + X_r$

$X_i \sim$ Geometric (p)

$E[X_i] = 1/p$, $Var(X_i) = (1-p)/p^2$

$$E[X] = r \cdot \left[\frac{1}{p}\right] = r/p$$

$$Var(X) = r \cdot \left[\frac{1-p}{p^2}\right] = \frac{r(1-p)}{p^2}$$

Eg: Find the expected value and variance of the number of times one must throw a die until the outcome 1 has occured 4 times.

→ RV here is sum of 4 rvs. $p = 1/6$

$X \sim$ neg Binomial $(r=4, p=1/6)$ $\quad X = X_1 + X_2 + X_3 + X_4$

$$E[X] = \frac{r}{p} = \frac{4}{1/6} = 24$$

$$Var[X] = \frac{r(1-p)}{p^2} = \frac{4 \cdot 5/6}{\frac{1}{6} \cdot \frac{1}{6}} = 120$$

Banach's match problem is a classic problem in probability attributed to Stefan Banach. Feller [1] says that the problem was inspired by a humorous reference to Banach's smoking habit in a speech honouring him by Hugo Steinhaus, but that it was not Banach who set the problem or provided an answer.

Suppose a mathematician carries two matchboxes at all times: one in his left pocket and one in his right. Each time he needs a match, he is equally likely to take it from either pocket. Suppose he reaches into his pocket and discovers for the first time that the box picked is empty. If it is assumed that each of the matchboxes originally contained N matches, what is the probability that there are exactly k matches in the other box?

Solution [edit]

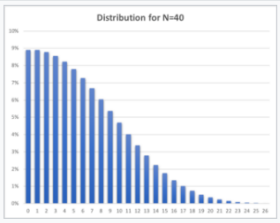
Without loss of generality consider the case where the matchbox in his right pocket has an unlimited number of matches and let M be the number of matches removed from this one before the left one is found to be empty. When the left pocket is found to be empty, the man has chosen that pocket $(N+1)$ times. Then M is the number of successes before $(N+1)$ failures in Bernoulli trials with $p = 1/2$, which has the negative binomial distribution and thus

$$P[M = m] = \binom{N+m}{m}\left(\frac{1}{2}\right)^{N+1+m}.$$

Returning to the original problem, we see that the probability that the left pocket is found to be empty first is $P[M < N+1]$ which equals $1/2$ because both are equally likely. We see that the number K of matches remaining in the other pocket is

$$P[K = k] = P[M = N-k | M < N+1] = 2P[M = N-k] = \binom{2N-k}{N-k}\left(\frac{1}{2}\right)^{2N-k}.$$

The expectation of the distribution is approximately $2\sqrt{N/\pi} - 1$. (This is shown using Stirling's approximation.[2]) So starting with boxes with $N = 40$ matches, the expected number of matches in the second box is 6.



Distribution of probability of having k matches remaining in the other pocket.

→ Initially, left N, right N

Say left is emptied first.

0 left, k right

N th trial for left; (N-k) th trial for right

Of trials gone = N + (N-k) = 2N-K

(2N-K+1) th trial to know that there's no more matches left.

when (N+1) th trial of left is made, it will be discovered that there's no matches left.

↳ = (N+1+N-K) th trial in total
= (2N-K+1) th trial

1 2 3 ... (2N-K trials) — Left 2N-K+1 th

N left, N-K right

Say left → success, right success.

So N+1 th success in 2N-K+1 th trial

X ~ Neg Binom(N+1, 1/2)

$P(E) = P(X = 2N-K+1) = {}^{2N-K}C_N \left(\frac{1}{2}\right)^{2N-K+1}$

$= {}^{2N-K}C_N \, p^N q^{2N-K-N} \cdot \frac{1}{2} = P(E)$

But right can also get empty at first.

$= 2 \cdot P(E)$

Hypergeometric random variable

Sums of Discrete Random Variables

Let X and Y be random variables. What is the distribution of X+Y?

If X and Y are not independent, we'd need the joint probabilities of X and Y to get the answer.
But if they are independent, we can get the distribution of X+Y without the probability.

By definition, Z can take all possible values of x+y where x and y are all possible values X and Y can take.

$$f_Z(z) = P\{Z=z\} = P\{X+Y=z\}$$
$$= \sum_{x+y=z} f(x,y) = \sum_x f(x, z-x)$$

eg.

Y \ X	0	1	2	3
-1	1/8	0	0	0
1	0	1/8	2/8	1/8
2	0	1/8	1/8	0
3	0	1/8	0	0

Joint PMF of X and Y
Find the PMF of X+Y

→ $X+Y : \{-1, 0, 1, 2, 3, 4, 5, 6\}$

$P\{X+Y=-1\} = P\{X=0, Y=-1\} = 1/8$
$P\{X+Y=0\} = P\{X=1, Y=-1\} = 0$
$P\{X+Y=1\} = P\{X=0, Y=1\} + P\{X=2, Y=-1\} = 0$
$P\{X+Y=2\} = 1/8$
$P\{X+Y=3\} = 3/8$
$P\{X+Y=4\} = 3/8$
$P\{X+Y=5\} = 0$
$P\{X+Y=6\} = 0$

x+y=z	-1	0	1	2	3	4	5	6
P(Z=z)	1/8	0	0	1/8	3/8	3/8	0	0

Sum of Independent Discrete Random Variables

If X and Y are independent DRVs, then the PMF of Z=X+Y can be calculated as

$$f_Z(z) = P\{X+Y=z\}$$
$$= \sum_x f_{X,Y}(x, z-x)$$
$$f_Z(z) = \sum_x f_X(x) \cdot f_Y(z-x)$$

→ convolution of discrete func"s.

as opposed to the continuous case $\int_{-\infty}^{\infty} f_X(x) \cdot f_Y(z-x)$

Sum of Independent Discrete Uniform Random Variables

Q. Let X and Y denote the outcomes of rolls of 2 dice. Find the PMF of X+Y.

→

x:	x	1	2	3	4	5	6
	$P_x(x)$	1/6	1/6	1/6	1/6	1/6	1/6

} same for y. X, Y are independent.

$X \sim U_{[1,6]}$

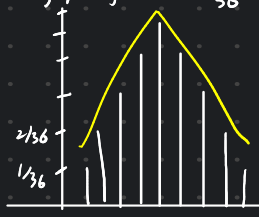
X can take 2, 3, 4, 5, ..., 12

$P\{X+Y=2\} = P\{X=1, Y=1\} = P\{X=1\} \cdot P\{Y=1\} = 1/6 \cdot 1/6 = 1/36$

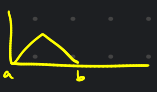
$P\{X+Y=4\} = P\{X=6, Y=3\} + P\{X=5, Y=4\} + P\{X=4, Y=5\} + P\{X=3, Y=6\} = 4 \cdot \frac{1}{36}$

and so on.

z	2	3	4	5	6	7	8	9	10	11	12
$P_Z(z)$	1/36	2/36	3/36	4/36	5/36	6/36	5/36	4/36	3/36	2/36	1/36



for continuous it was like



Sum of Independent Discrete Bernoulli Random Variables

Q. Let X and Y represent independent Bernoulli variables with same probability of success p.
What is the PMF of X+Y?

→

X: {0, 1} (1-p) p

Y: {0, 1} (1-p) p

$Z = X+Y : \{0, 1, 2\}$

$P\{Z=0\} = P\{X=0, Y=0\} = (1-p)^2$

$P\{Z=1\} = 2 \cdot p(1-p)$

$P\{Z=2\} = p^2$

$P(Z=z) = \binom{2}{z} p^z (1-p)^{2-z}$

Ug, on adding n independent Bernoulli rvs,

$P(Z=z) = P\{X_1+X_2+\dots+X_n = z\} = \binom{n}{z} p^z (1-p)^{n-z}$

$Z \sim \text{Binom}(n, p)$

∴ sum of independent Bernoulli variables gives a Binomial rv.

Sum of Independent Discrete Binomial Random Variables

Q. Let X and Y represent independent Binomial variables.

$X \sim \text{Binom}(n, p)$

$Y \sim \text{Binom}(m, p)$

What is the PMF of X+Y?

→ X — sum of n Bernoulli(p) vars. $X = X_1 + \dots + X_n$

Y — " m " " $Y = Y_1 + \dots + Y_m$

$Z = X+Y$ sum of m+n Bernoulli(p) vars. = Binom(m+n, p) $P_Z(z) = \binom{m+n}{z} p^z (1-p)^{m+n-z}$

OR $Z = X_1 + \dots + X_n + Y_1 + \dots + Y_m$

$f_Z(z) = P\{X+Y=z\}$

$= \sum_{x=0}^{n} P_X(x) \cdot P_Y(z-x)$

$= \sum_{x=0}^{n} \binom{n}{x} p^x (1-p)^{n-x} \binom{m}{z-x} p^{z-x} (1-p)^{m-z+x}$

$= \sum_{x=0}^{n} \binom{n}{x}\binom{m}{z-x} p^z (1-p)^{n+m-z}$

$= p^z (1-p)^{m+n-z} \sum_{x=0}^{n} \binom{n}{x}\binom{m}{z-x} = \binom{m+n}{z} p^z (1-p)^{m+n-z} \sim \text{Binom}(m+n, p)$

sum of 2 binomials is a binomial IF the prob of SUCCESS is same for both.

Same for multiple rvs, not just 2.

Sum of Independent Poisson Random Variables

Let X and Y be independent Poisson rvs. Find PMF of Z=X+Y.

$X \sim \text{Pois}(\lambda_1)$, $Y \sim \text{Pois}(\lambda_2)$

→

$$P_Z(z) = P\{X+Y=z\}$$

$$= \sum_x f_X(x) \cdot f_Y(z-x)$$

$$= \sum_{x=0}^{z} e^{-\lambda_1} \frac{\lambda_1^x}{x!} \cdot e^{-\lambda_2} \frac{\lambda_2^{(z-x)}}{(z-x)!}$$

$$= e^{-(\lambda_1+\lambda_2)} \sum_{x=0}^{z} \frac{1}{x!(z-x)!} (\lambda_1)^x (\lambda_2)^{z-x}$$

$$= \frac{e^{-(\lambda_1+\lambda_2)}}{z!} \sum_{x=0}^{z} \frac{z!}{x!(z-x)!} \lambda_1^x \lambda_2^{(z-x)}$$

$$= \frac{e^{-(\lambda_1+\lambda_2)} (\lambda_1+\lambda_2)^z}{z!} \sim \text{Pois}(\lambda_1+\lambda_2)$$

$$(\lambda_1+\lambda_2)^z = \sum_{x=0}^{z} \binom{z}{x} (\lambda_1)^x (\lambda_2)^{z-x}$$

$$= \sum_{x=0}^{z} \frac{z!}{x!(z-x)!} \lambda_1^x \lambda_2^{(z-x)}$$

Sum of n independent Poisson rvs with parameters λ1, λ2, ..., λn is a Poisson rv with parameter (λ1 + ... + λn).

Sum of Independent Geometric Random Variables

Say X and Y are geometric rvs with probability of success p (# of Bernoulli trials reqd to get a success).

$X \sim \text{Geometric}(p)$; $Y \sim \text{Geometric}(p)$.

$P_X(x) = (1-p)^{x-1} p^1$; $P_Y(y) = (1-p)^{y-1} p$

$Z = X+Y$

$$P_Z(z) = \sum_{x=1}^{z-1} P_X(x) \cdot P_Y(z-x)$$

$$= \sum_{x=1}^{z-1} (1-p)^{x-1} \cdot p \cdot (1-p)^{z-x-1} \cdot p$$

$$= \sum_{x=1}^{z-1} (1-p)^{z-2} p^2$$

$$= (1-p)^{z-2} p^2 \sum_{x=1}^{z-1} 1$$

$P_Z(z) = (z-1) \cdot p^2 \cdot (1-p)^{z-2}$ } Similar to PMF of -ve Binomial rar (# of trials to get r successes)

$\sim$ Neg. Binom (2, p)

where PMF $= \binom{x-1}{r-1} p^r (1-p)^{x-r}$

For r=2, $\binom{x-1}{1} p^2 (1-p)^{x-2}$

$= (x-1) p^2 (1-p)^{x-2}$

Sum of n independent geometric rvs (p) is a -ve Binomial rv with parameters (r=n, p)

PMF: $\binom{x-1}{n-1} p^n (1-p)^{x-n}$

Sum of Independent Negative Binomial Random Variables

$X \sim$ Negative Binomial (r_1, p) = sum of r_1 independent -ve Binomial rvs } same prob. of success.

$Y \sim$ Negative Binomial (r_2, p) = sum of r_2 ind. -ve Binom. rvs.

$Z = X+Y$

Z = sum of $r_1 + r_2$ ind. rvs with parameter p = -ve Binom. with parameters $(r = r_1+r_2, p)$

$Z \sim$ Neg. Binom (r_1+r_2, p)

PMF = $\binom{x-1}{r_1+r_2-1} p^{r_1+r_2} (1-p)^{x-(r_1+r_2)}$